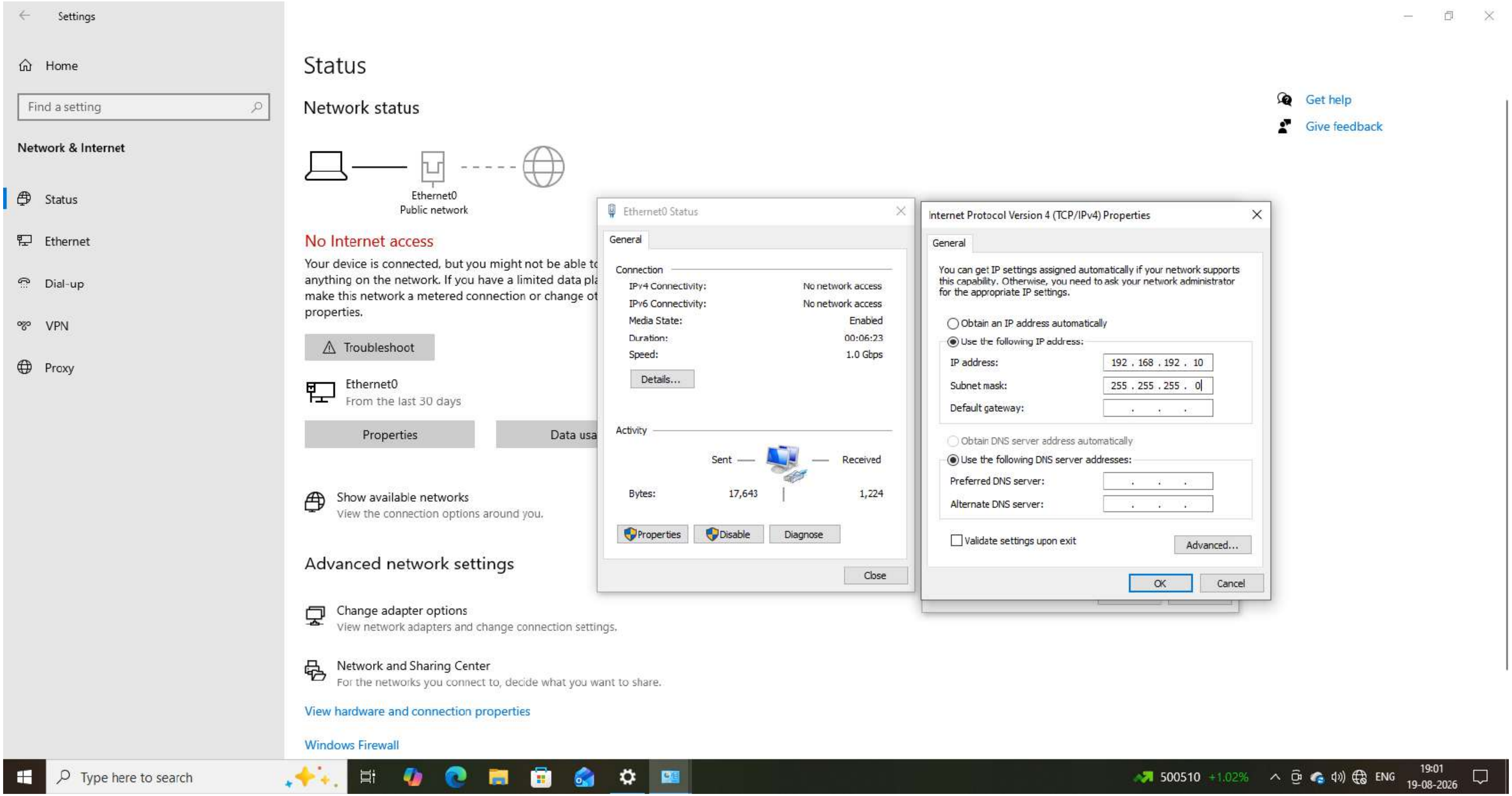


Settings
Home
Find a setting
Network & Internet
Status
Ethernet
Dial-up
VPN
Proxy
Status
Network status
Ethernet0
Public network
No Internet access
Your device is connected, but you might not be able to
anything on the network. If you have a limited data pla
make this network a metered connection or change ot
properties.
Troubleshoot
Ethernet0
From the last 30 days
Properties
Data usa
Show available networks
View the connection options around you.
Advanced network settings
Change adapter options
View network adapters and change connection settings.
Network and Sharing Center
For the networks you connect to, decide what you want to share.
View hardware and connection properties
Windows Firewall
Get help
Give feedback
Ethernet0 Status
General
Connection
IPv4 Connectivity: No network access
IPv6 Connectivity: No network access
Media State: Enabled
Duration: 00:06:23
Speed: 1.0 Gbps
Details...
Activity
Sent
Received
Bytes: 17,643 1,224
Properties
Disable
Diagnose
Close
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 192 . 168 . 192 . 10
Subnet mask: 255 . 255 . 255 . 0
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...
OK
Cancel
Type here to search
500510 +1.02%
ENG
19:01
19-08-2026

C:\Windows\system32\CMD.exe

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\riken>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::41d6:94cb:ff32:5ab4%5
   IPv4 Address. . . . . . . . . . . : 192.168.192.10
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

C:\Users\riken>\
```

Type here to search

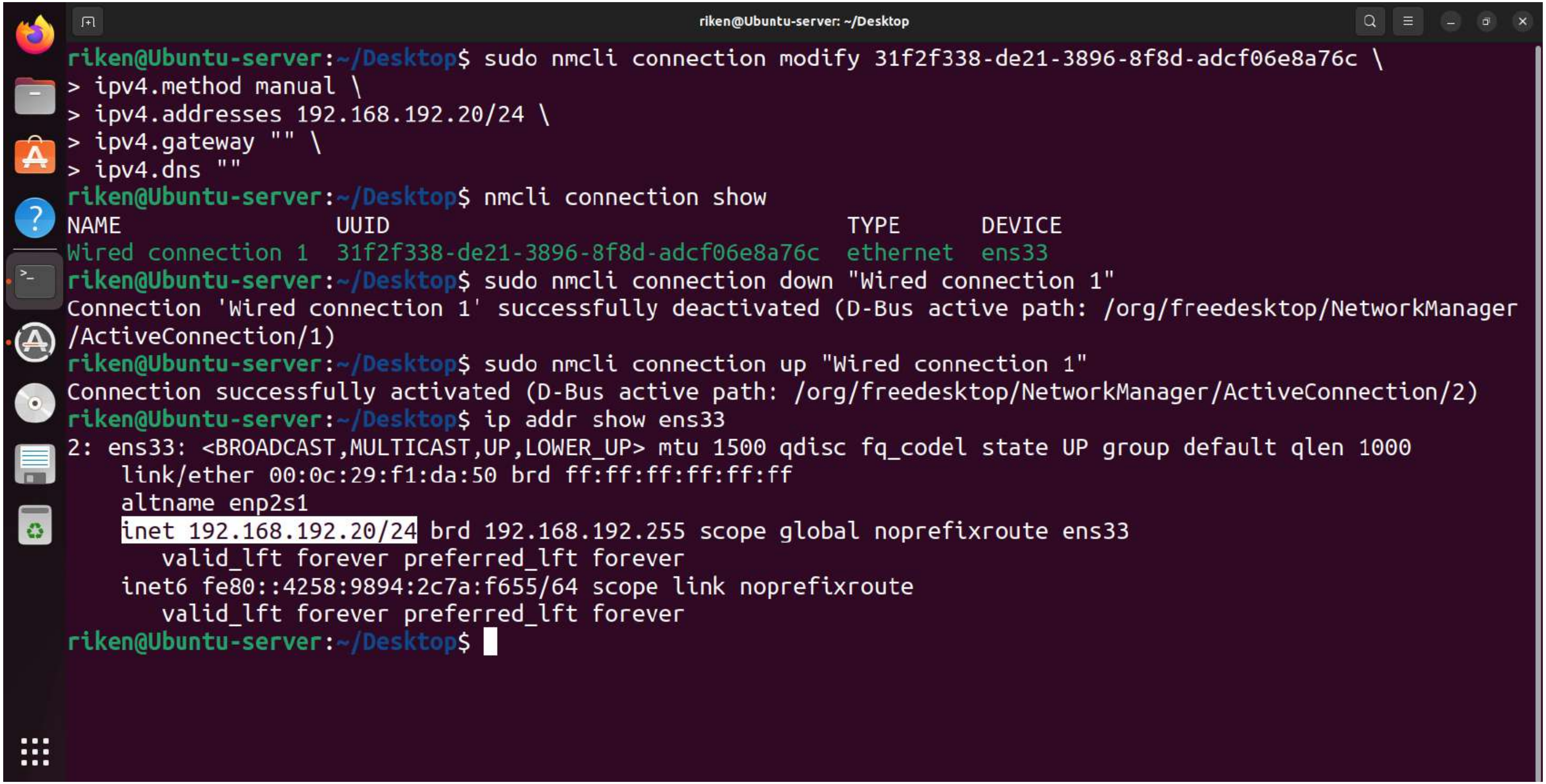
500875 +1.16% ENG 19:03 19-08-2026

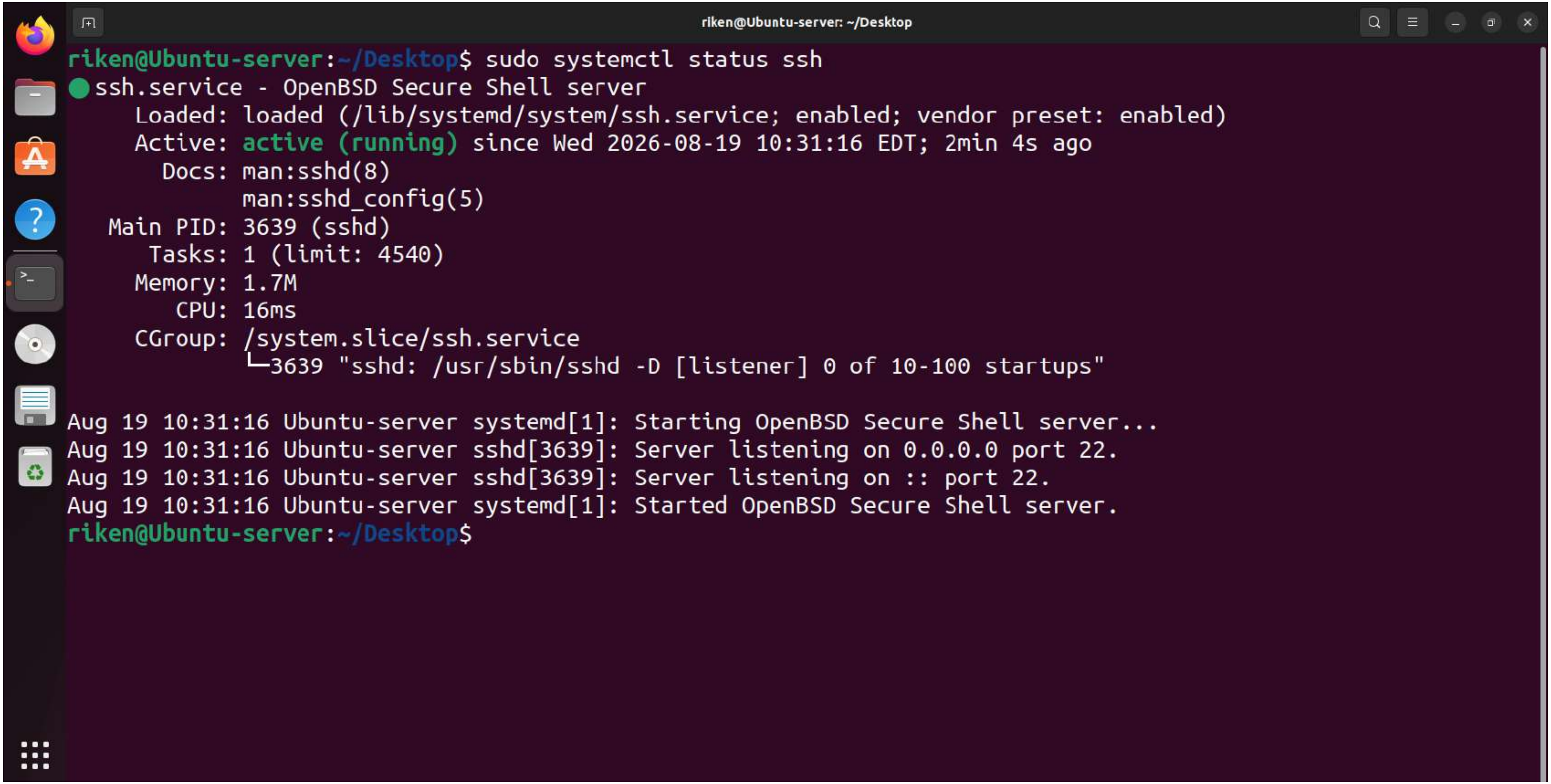
C:\Windows\system32\CMD.exe

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\riken>ping 192.168.192.20

Pinging 192.168.192.20 with 32 bytes of data:
Reply from 192.168.192.20: bytes=32 time<1ms TTL=64
Reply from 192.168.192.20: bytes=32 time<1ms TTL=64
Reply from 192.168.192.20: bytes=32 time<1ms TTL=64
Reply from 192.168.192.20: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.192.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\riken>
```

Type here to search

26°C Cloudy ENG 19:37 19-08-2026


riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ sudo nmcli connection modify 31f2f338-de21-3896-8f8d-adcf06e8a76c \
> ipv4.method manual \
> ipv4.addresses 192.168.192.20/24 \
> ipv4.gateway "" \
> ipv4.dns ""
riken@Ubuntu-server:~/Desktop$ nmcli connection show
NAME UUID TYPE DEVICE
Wired connection 1 31f2f338-de21-3896-8f8d-adcf06e8a76c ethernet ens33
riken@Ubuntu-server:~/Desktop$ sudo nmcli connection down "Wired connection 1"
Connection 'Wired connection 1' successfully deactivated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/1)
riken@Ubuntu-server:~/Desktop$ sudo nmcli connection up "Wired connection 1"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/2)
riken@Ubuntu-server:~/Desktop$ ip addr show ens33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 00:0c:29:f1:da:50 brd ff:ff:ff:ff:ff:ff
altname enp2s1
inet 192.168.192.20/24 brd 192.168.192.255 scope global noprefixroute ens33
valid_lft forever preferred_lft forever
inet6 fe80::4258:9894:2c7a:f655/64 scope link noprefixroute
valid_lft forever preferred_lft forever
riken@Ubuntu-server:~/Desktop$


riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ sudo systemctl status ssh
ssh.service - OpenBSD Secure Shell server
Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
Active: active (running) since Wed 2026-08-19 10:31:16 EDT; 2min 4s ago
Docs: man:sshd(8)
man:sshd_config(5)
Main PID: 3639 (sshd)
Tasks: 1 (limit: 4540)
Memory: 1.7M
CPU: 16ms
CGroup: /system.slice/ssh.service
└─3639 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
Aug 19 10:31:16 Ubuntu-server systemd[1]: Starting OpenBSD Secure Shell server...
Aug 19 10:31:16 Ubuntu-server sshd[3639]: Server listening on 0.0.0.0 port 22.
Aug 19 10:31:16 Ubuntu-server sshd[3639]: Server listening on :: port 22.
Aug 19 10:31:16 Ubuntu-server systemd[1]: Started OpenBSD Secure Shell server.
riken@Ubuntu-server:~/Desktop$

adminuser@Ubuntu-server: ~

```
PS C:\Users\riken> Get-Content "$env:USERPROFILE\.ssh\id_ed25519.pub" | ssh adminuser@192.168.192.20 "mkdir -p ~/.ssh && chmod 700 ~/.ssh &&
cat >> ~/.ssh/authorized_keys && chmod 600 ~/.ssh/authorized_keys"
adminuser@192.168.192.20's password:
PS C:\Users\riken> ssh adminuser@192.168.192.20
Enter passphrase for key 'C:\Users\riken/.ssh/id_ed25519':
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-111-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

197 updates can be applied immediately.
177 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

adminuser@Ubuntu-server:~$
```

Type here to search

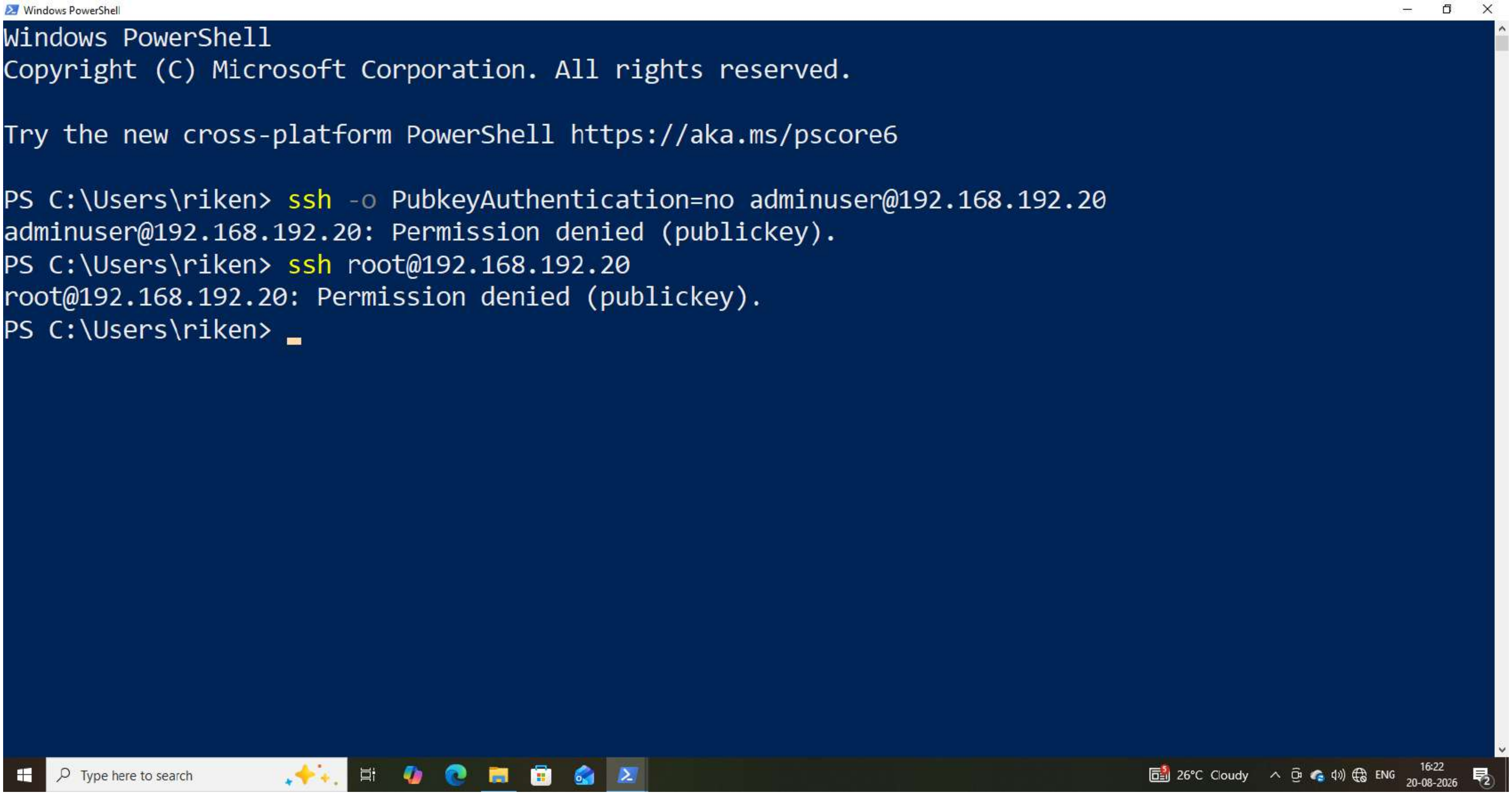
26°C Cloudy ENG 15:40 20-08-2026

adminuser@Ubuntu-server: ~

```
  GNU nano 6.2                          /etc/ssh/sshd_config *

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 1

PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile     .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes
# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication no
#PermitEmptyPasswords no

^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark  M-] To Bracket
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy      ^Q Where Was
```

Type here to search    26°C Cloudy    ENG    15:59 20-08-2026


riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ sudo sshd -T | grep -E 'permitrootlogin|passwordauthentication|pubkeyauthentication'
permitrootlogin no
pubkeyauthentication yes
passwordauthentication no
riken@Ubuntu-server:~/Desktop$

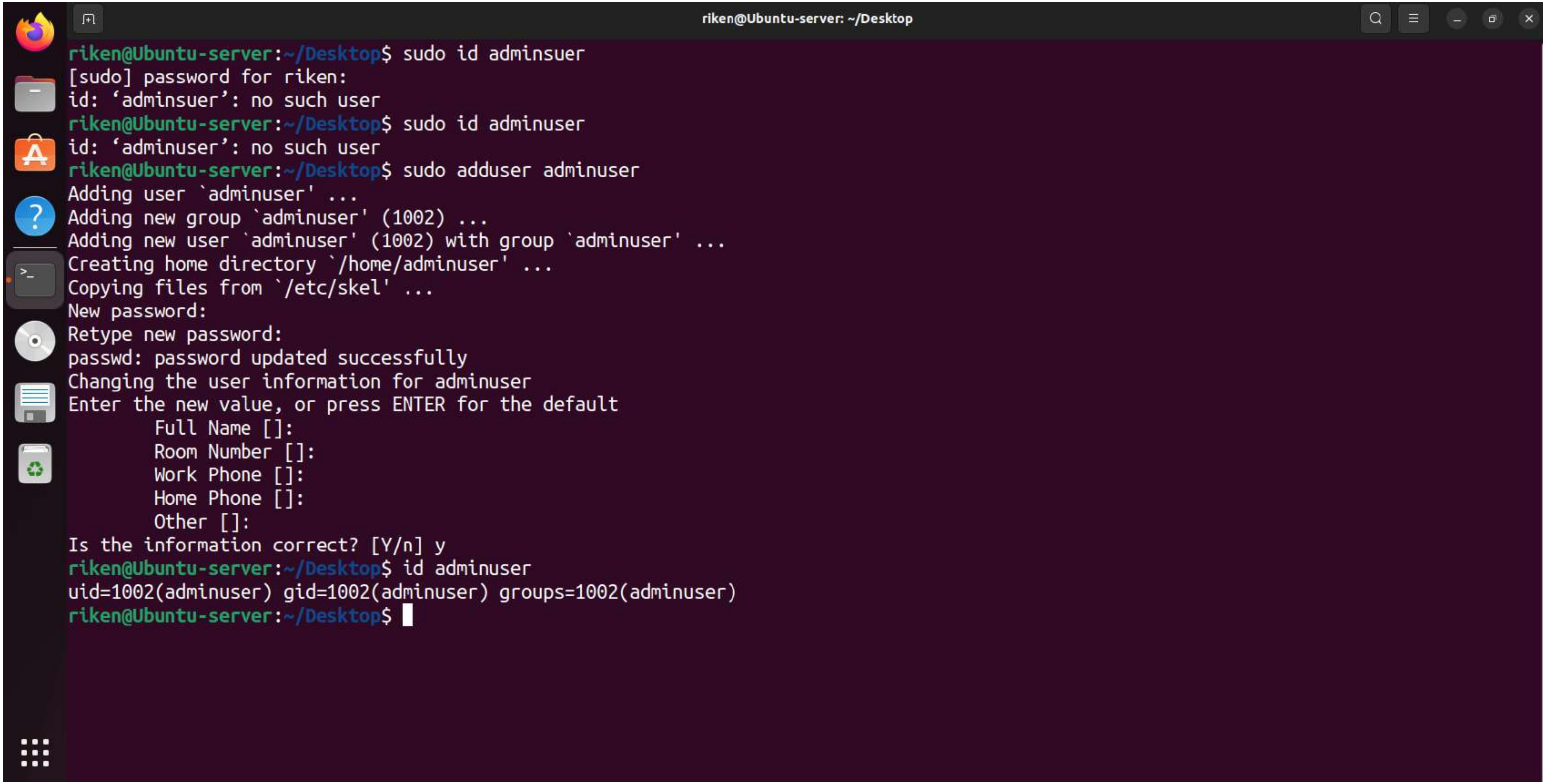

Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Try the new cross-platform PowerShell https://aka.ms/pscore6
PS C:\Users\riken> ssh -o PubkeyAuthentication=no adminuser@192.168.192.20
adminuser@192.168.192.20: Permission denied (publickey).
PS C:\Users\riken> ssh root@192.168.192.20
root@192.168.192.20: Permission denied (publickey).
PS C:\Users\riken>
Type here to search
26°C Cloudy
ENG
16:22
20-08-2026

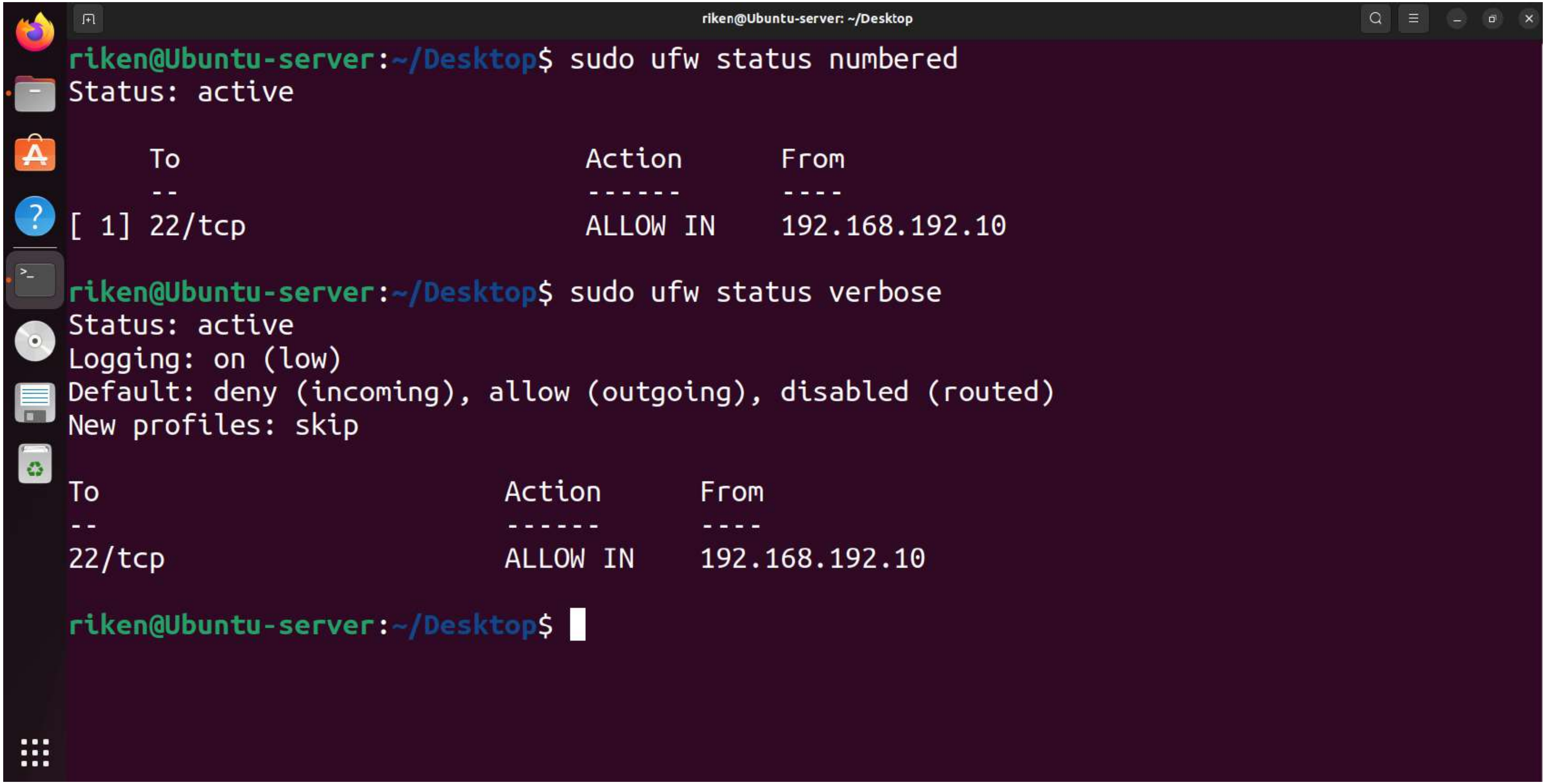
adminuser@Ubuntu-server: ~

```
adminuser@Ubuntu-server:~$ adminuser@Ubuntu-server:~$ sudo sshd -t
adminuser@Ubuntu-server:~$ sudo systemctl restart ssh
adminuser@Ubuntu-server:~$ sudo systemctl satatus ssh --no-pager
Unknown command verb satatus.
adminuser@Ubuntu-server:~$ sudo systemctl status ssh --no-pager
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-08-20 06:36:12 EDT; 54s ago
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 6805 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 6807 (sshd)
      Tasks: 1 (limit: 4540)
     Memory: 1.7M
        CPU: 14ms
     CGroup: /system.slice/ssh.service
             └─6807 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 20 06:36:12 Ubuntu-server systemd[1]: Starting OpenBSD Secure Shell server...
Aug 20 06:36:12 Ubuntu-server sshd[6807]: Server listening on 0.0.0.0 port 22.
Aug 20 06:36:12 Ubuntu-server sshd[6807]: Server listening on :: port 22.
Aug 20 06:36:12 Ubuntu-server systemd[1]: Started OpenBSD Secure Shell server.
adminuser@Ubuntu-server:~$ _
```

Type here to search

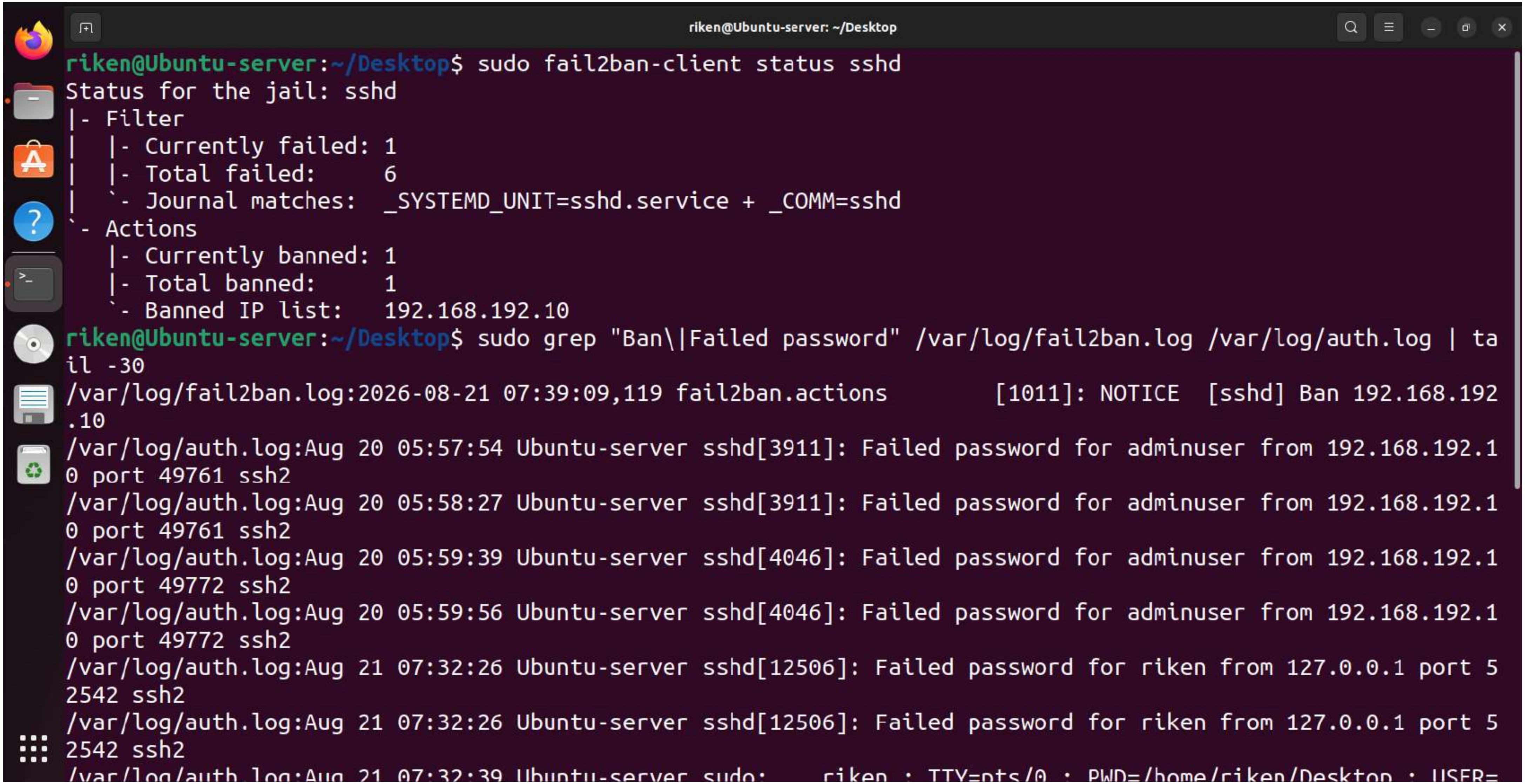
26°C Cloudy ENG 16:08 20-08-2026


riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ sudo id adminsuer
[sudo] password for riken:
id: ‘adminsuer’: no such user
riken@Ubuntu-server:~/Desktop$ sudo id adminuser
id: ‘adminuser’: no such user
riken@Ubuntu-server:~/Desktop$ sudo adduser adminuser
Adding user `adminuser' ...
Adding new group `adminuser' (1002) ...
Adding new user `adminuser' (1002) with group `adminuser' ...
Creating home directory `/home/adminuser' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for adminuser
Enter the new value, or press ENTER for the default
Full Name []:
Room Number []:
Work Phone []:
Home Phone []:
Other []:
Is the information correct? [Y/n] y
riken@Ubuntu-server:~/Desktop$ id adminuser
uid=1002(adminuser) gid=1002(adminuser) groups=1002(adminuser)
riken@Ubuntu-server:~/Desktop$

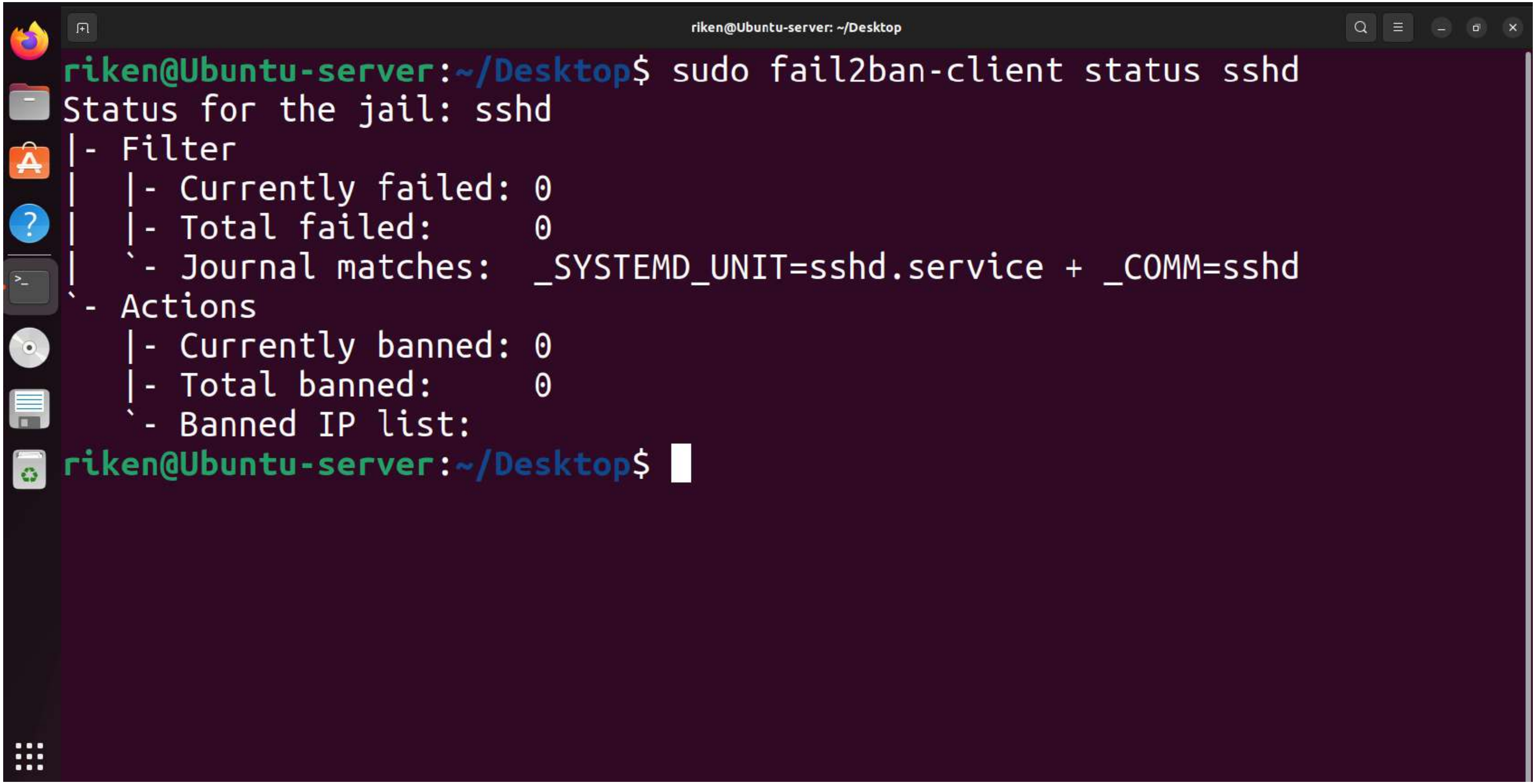

riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ sudo ufw status numbered
Status: active
To Action From
-- ------ ----
[ 1] 22/tcp ALLOW IN 192.168.192.10
riken@Ubuntu-server:~/Desktop$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
To Action From
-- ------ ----
22/tcp ALLOW IN 192.168.192.10
riken@Ubuntu-server:~/Desktop$

riken@Ubuntu-server: ~/Desktop

```
Preparing to unpack .../whois_5.5.13_amd64.deb ...
Unpacking whois (5.5.13) ...
Setting up whois (5.5.13) ...
Setting up fail2ban (0.11.2-6) ...
Setting up python3-pyinotify (0.9.6-1.3) ...
Processing triggers for man-db (2.10.2-1) ...
riken@Ubuntu-server:~/Desktop$ sudo systemctl enable --now fail2ban
Synchronizing state of fail2ban.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable fail2ban
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /lib/systemd/system/fail2ban.
service.
riken@Ubuntu-server:~/Desktop$ sudo systemctl status fail2ban
● fail2ban.service - Fail2Ban Service
     Loaded: loaded (/lib/systemd/system/fail2ban.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2026-08-20 07:26:16 EDT; 22s ago
       Docs: man:fail2ban(1)
   Main PID: 4404 (fail2ban-server)
      Tasks: 5 (limit: 4540)
     Memory: 12.4M
        CPU: 109ms
     CGroup: /system.slice/fail2ban.service
             └─4404 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Aug 20 07:26:16 Ubuntu-server systemd[1]: Started Fail2Ban Service.
Aug 20 07:26:16 Ubuntu-server fail2ban-server[4404]: Server ready
riken@Ubuntu-server:~/Desktop$
```



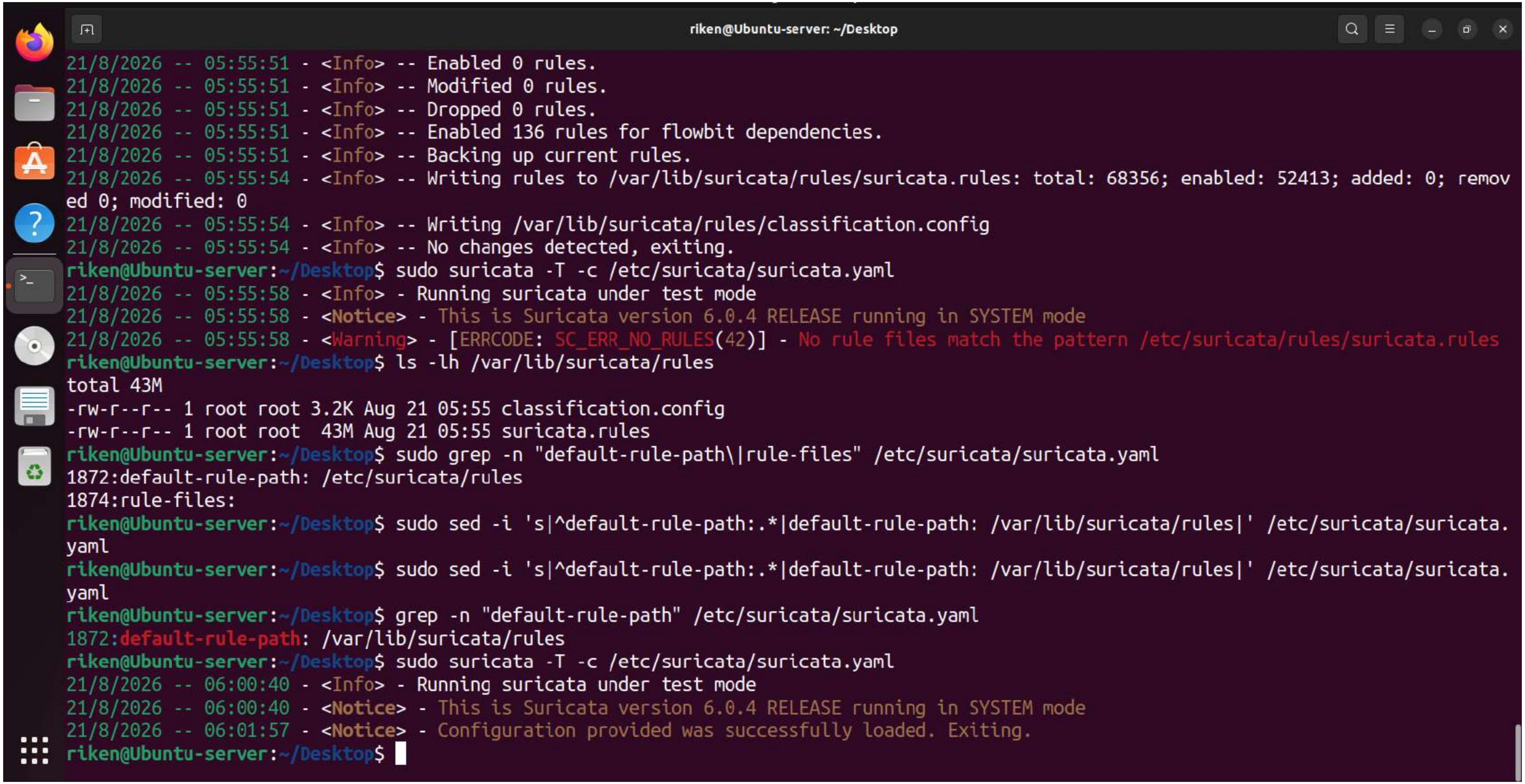
```
riken@Ubuntu-server:~/Desktop$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 1
|  |- Total failed:     6
|  `- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd
`- Actions
   |- Currently banned: 1
   |- Total banned:     1
   `- Banned IP list:   192.168.192.10
riken@Ubuntu-server:~/Desktop$ sudo grep "Ban\|Failed password" /var/log/fail2ban.log /var/log/auth.log | ta
il -30
/var/log/fail2ban.log:2026-08-21 07:39:09,119 fail2ban.actions        [1011]: NOTICE  [sshd] Ban 192.168.192
.10
/var/log/auth.log:Aug 20 05:57:54 Ubuntu-server sshd[3911]: Failed password for adminuser from 192.168.192.1
0 port 49761 ssh2
/var/log/auth.log:Aug 20 05:58:27 Ubuntu-server sshd[3911]: Failed password for adminuser from 192.168.192.1
0 port 49761 ssh2
/var/log/auth.log:Aug 20 05:59:39 Ubuntu-server sshd[4046]: Failed password for adminuser from 192.168.192.1
0 port 49772 ssh2
/var/log/auth.log:Aug 20 05:59:56 Ubuntu-server sshd[4046]: Failed password for adminuser from 192.168.192.1
0 port 49772 ssh2
/var/log/auth.log:Aug 21 07:32:26 Ubuntu-server sshd[12506]: Failed password for riken from 127.0.0.1 port 5
2542 ssh2
/var/log/auth.log:Aug 21 07:32:26 Ubuntu-server sshd[12506]: Failed password for riken from 127.0.0.1 port 5
2542 ssh2
/var/log/auth.log:Aug 21 07:32:39 Ubuntu-server sudo:    riken : TTY=pts/0 ; PWD=/home/riken/Desktop ; USER=
```


riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ sudo fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  `- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:
riken@Ubuntu-server:~/Desktop$

riken@Ubuntu-server: ~/Desktop

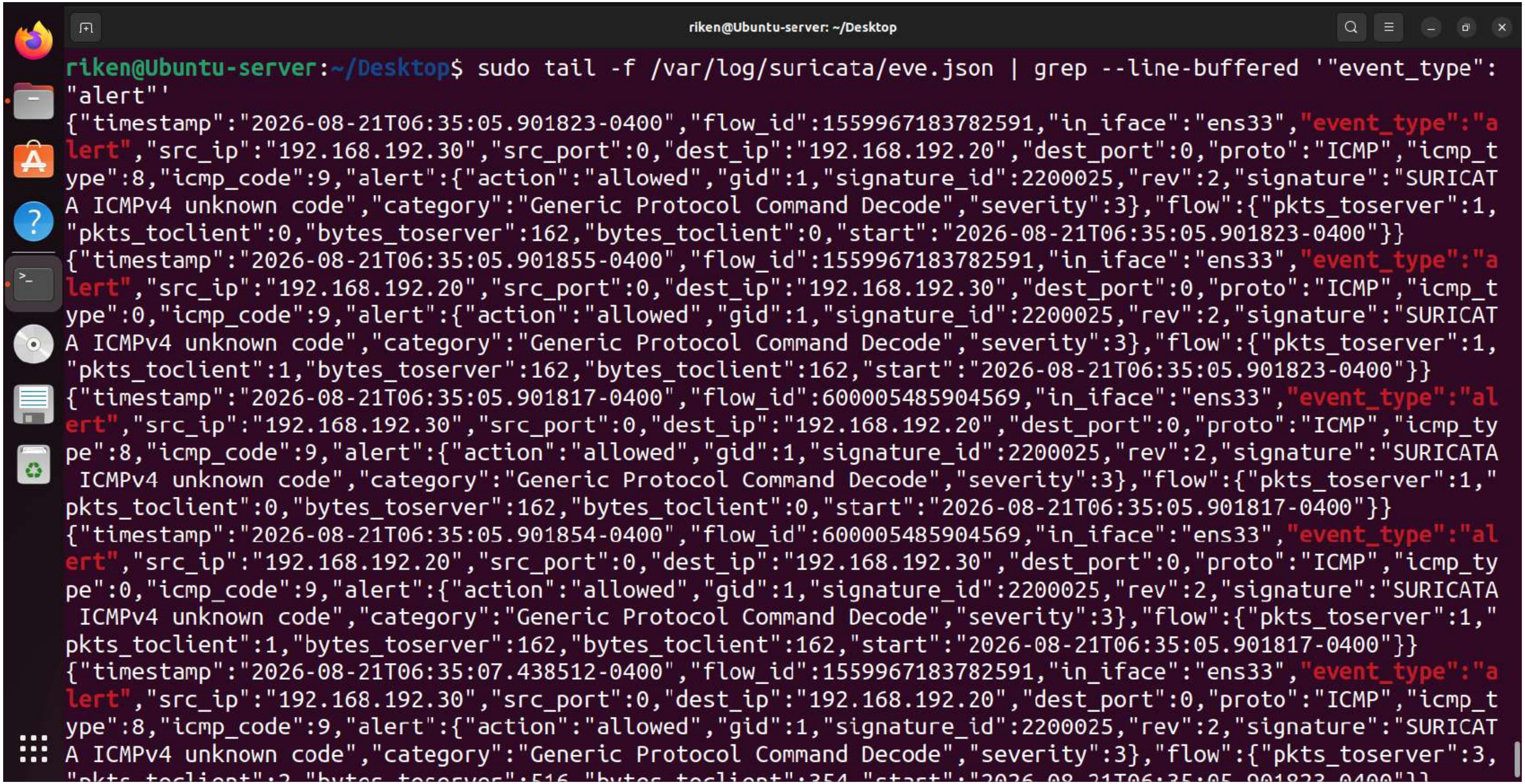
```
riken@Ubuntu-server:~/Desktop$ sudo chmod 750 /srv/meridian-data
riken@Ubuntu-server:~/Desktop$ ls -ld /srv/meridian-data
drwxr-x--- 2 adminuser adminuser 4096 Aug 20 10:19 /srv/meridian-data
riken@Ubuntu-server:~/Desktop$ sudo -u adminuser touch /srv/meridian-data/testfile.txt
riken@Ubuntu-server:~/Desktop$ ls -l /srv/meridian-data
ls: cannot open directory '/srv/meridian-data': Permission denied
riken@Ubuntu-server:~/Desktop$ sudo ls -l /srv/meridian-data
total 0
-rw-rw-r-- 1 adminuser adminuser 0 Aug 20 10:34 testfile.txt
riken@Ubuntu-server:~/Desktop$ ls -l /etc/passwd /etc/shadow
-rw-r--r-- 1 root root   3046 Aug 20 05:41 /etc/passwd
-rw-r----- 1 root shadow 1621 Aug 20 05:41 /etc/shadow
riken@Ubuntu-server:~/Desktop$ sudo -u adminuser cat /etc/sh
shadow   shadow-  shells
riken@Ubuntu-server:~/Desktop$ sudo -u adminuser cat /etc/sh
shadow   shadow-  shells
riken@Ubuntu-server:~/Desktop$ sudo -u adminuser cat /etc/sh
shadow   shadow-  shells
riken@Ubuntu-server:~/Desktop$ sudo -u adminuser cat /etc/shadow
cat: /etc/shadow: Permission denied
riken@Ubuntu-server:~/Desktop$ sudo -u adminuser cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
```

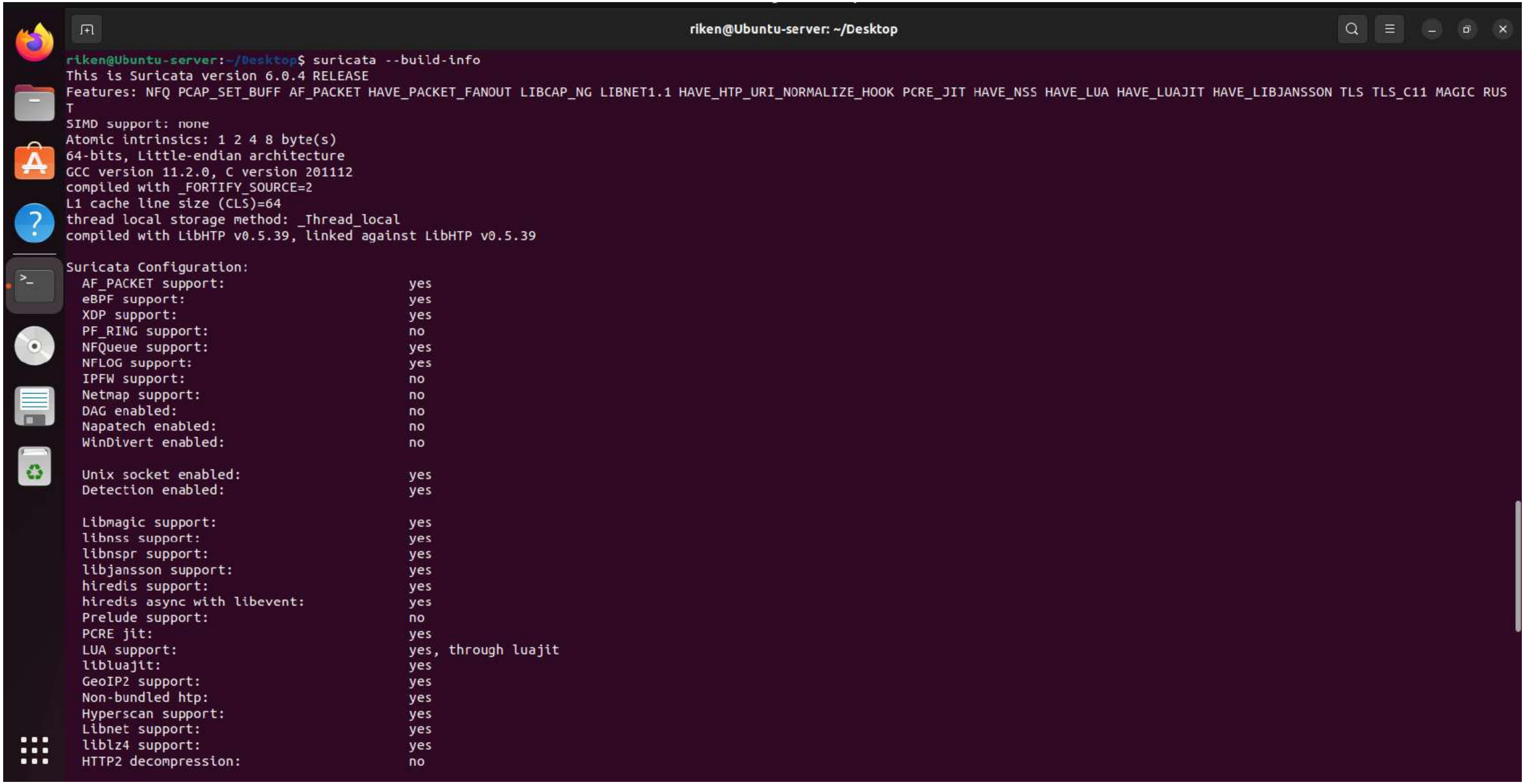

riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ sudo grep -R "pam_pwquality\|minlen\|minclass" /etc/pam.d/common-password /etc/security/pwquality.conf
/etc/pam.d/common-password:password requisite pam_pwquality.so retry=3
/etc/security/pwquality.conf:# minlen = 8
/etc/security/pwquality.conf:# minclass = 0
riken@Ubuntu-server:~/Desktop$ sudo nano /etc/se
security/ selinux/ sensors3.conf sensors.d/ services
riken@Ubuntu-server:~/Desktop$ sudo nano /etc/se
security/ selinux/ sensors3.conf sensors.d/ services
riken@Ubuntu-server:~/Desktop$ sudo nano /etc/se
security/ selinux/ sensors3.conf sensors.d/ services
riken@Ubuntu-server:~/Desktop$ sudo nano /etc/security/pwquality.conf
riken@Ubuntu-server:~/Desktop$ sudo nano /etc/security/pwquality.conf
riken@Ubuntu-server:~/Desktop$ grep -E '^(minlen|maxlen|minclass)' /etc/security/pwquality.conf
minlen = 8
maxlen = 52
minclass = 4
riken@Ubuntu-server:~/Desktop$

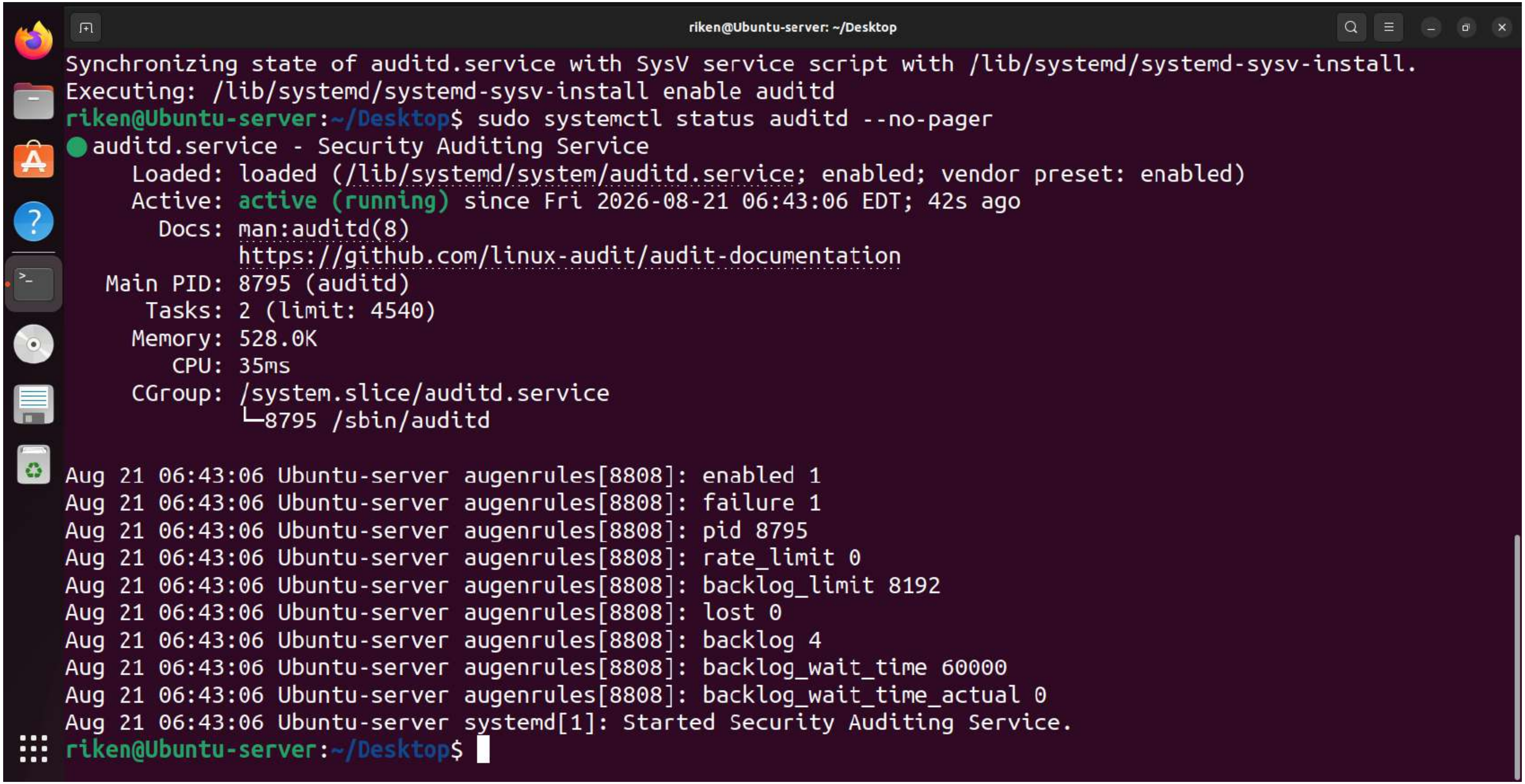

riken@Ubuntu-server: ~/Desktop
21/8/2026 -- 05:55:51 - <Info> -- Enabled 0 rules.
21/8/2026 -- 05:55:51 - <Info> -- Modified 0 rules.
21/8/2026 -- 05:55:51 - <Info> -- Dropped 0 rules.
21/8/2026 -- 05:55:51 - <Info> -- Enabled 136 rules for flowbit dependencies.
21/8/2026 -- 05:55:51 - <Info> -- Backing up current rules.
21/8/2026 -- 05:55:54 - <Info> -- Writing rules to /var/lib/suricata/rules/suricata.rules: total: 68356; enabled: 52413; added: 0; removed 0; modified: 0
21/8/2026 -- 05:55:54 - <Info> -- Writing /var/lib/suricata/rules/classification.config
21/8/2026 -- 05:55:54 - <Info> -- No changes detected, exiting.
riken@Ubuntu-server:~/Desktop$ sudo suricata -T -c /etc/suricata/suricata.yaml
21/8/2026 -- 05:55:58 - <Info> - Running suricata under test mode
21/8/2026 -- 05:55:58 - <Notice> - This is Suricata version 6.0.4 RELEASE running in SYSTEM mode
21/8/2026 -- 05:55:58 - <Warning> - [ERRCODE: SC_ERR_NO_RULES(42)] - No rule files match the pattern /etc/suricata/rules/suricata.rules
riken@Ubuntu-server:~/Desktop$ ls -lh /var/lib/suricata/rules
total 43M
-rw-r--r-- 1 root root 3.2K Aug 21 05:55 classification.config
-rw-r--r-- 1 root root 43M Aug 21 05:55 suricata.rules
riken@Ubuntu-server:~/Desktop$ sudo grep -n "default-rule-path\|rule-files" /etc/suricata/suricata.yaml
1872:default-rule-path: /etc/suricata/rules
1874:rule-files:
riken@Ubuntu-server:~/Desktop$ sudo sed -i 's|^default-rule-path:.*|default-rule-path: /var/lib/suricata/rules|' /etc/suricata/suricata.yaml
riken@Ubuntu-server:~/Desktop$ sudo sed -i 's|^default-rule-path:.*|default-rule-path: /var/lib/suricata/rules|' /etc/suricata/suricata.yaml
riken@Ubuntu-server:~/Desktop$ grep -n "default-rule-path" /etc/suricata/suricata.yaml
1872:default-rule-path: /var/lib/suricata/rules
riken@Ubuntu-server:~/Desktop$ sudo suricata -T -c /etc/suricata/suricata.yaml
21/8/2026 -- 06:00:40 - <Info> - Running suricata under test mode
21/8/2026 -- 06:00:40 - <Notice> - This is Suricata version 6.0.4 RELEASE running in SYSTEM mode
21/8/2026 -- 06:01:57 - <Notice> - Configuration provided was successfully loaded. Exiting.
riken@Ubuntu-server:~/Desktop$

riken@Ubuntu-server: ~/Desktop

```
riken@Ubuntu-server:~/Desktop$ sudo systemctl restart suricata
Job for suricata.service failed because the control process exited with error code.
See "systemctl status suricata.service" and "journalctl -xeu suricata.service" for details.
riken@Ubuntu-server:~/Desktop$ sudo systemctl status suricata --no-pager
● suricata.service - Suricata IDS/IDP daemon
     Loaded: loaded (/lib/systemd/system/suricata.service; enabled; vendor preset: enabled)
     Active: active (running) since Fri 2026-08-21 06:20:40 EDT; 36s ago
       Docs: man:suricata(8)
             man:suricatasc(8)
             https://suricata-ids.org/docs/
    Process: 6634 ExecStart=/usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/s
uricata.pid (code=exited, status=0/SUCCESS)
   Main PID: 6635 (Suricata-Main)
      Tasks: 1 (limit: 4540)
     Memory: 596.3M
        CPU: 36.242s
     CGroup: /system.slice/suricata.service
             └─6635 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /run/suricata.
pid

Aug 21 06:20:40 Ubuntu-server systemd[1]: Starting Suricata IDS/IDP daemon...
Aug 21 06:20:40 Ubuntu-server suricata[6634]: 21/8/2026 -- 06:20:40 - <Notice> - This is Suricata version 6.
0.4 RELEASE runnin…STEM mode
Aug 21 06:20:40 Ubuntu-server systemd[1]: Started Suricata IDS/IDP daemon.
Hint: Some lines were ellipsized, use -l to show in full.
riken@Ubuntu-server:~/Desktop$ 
```

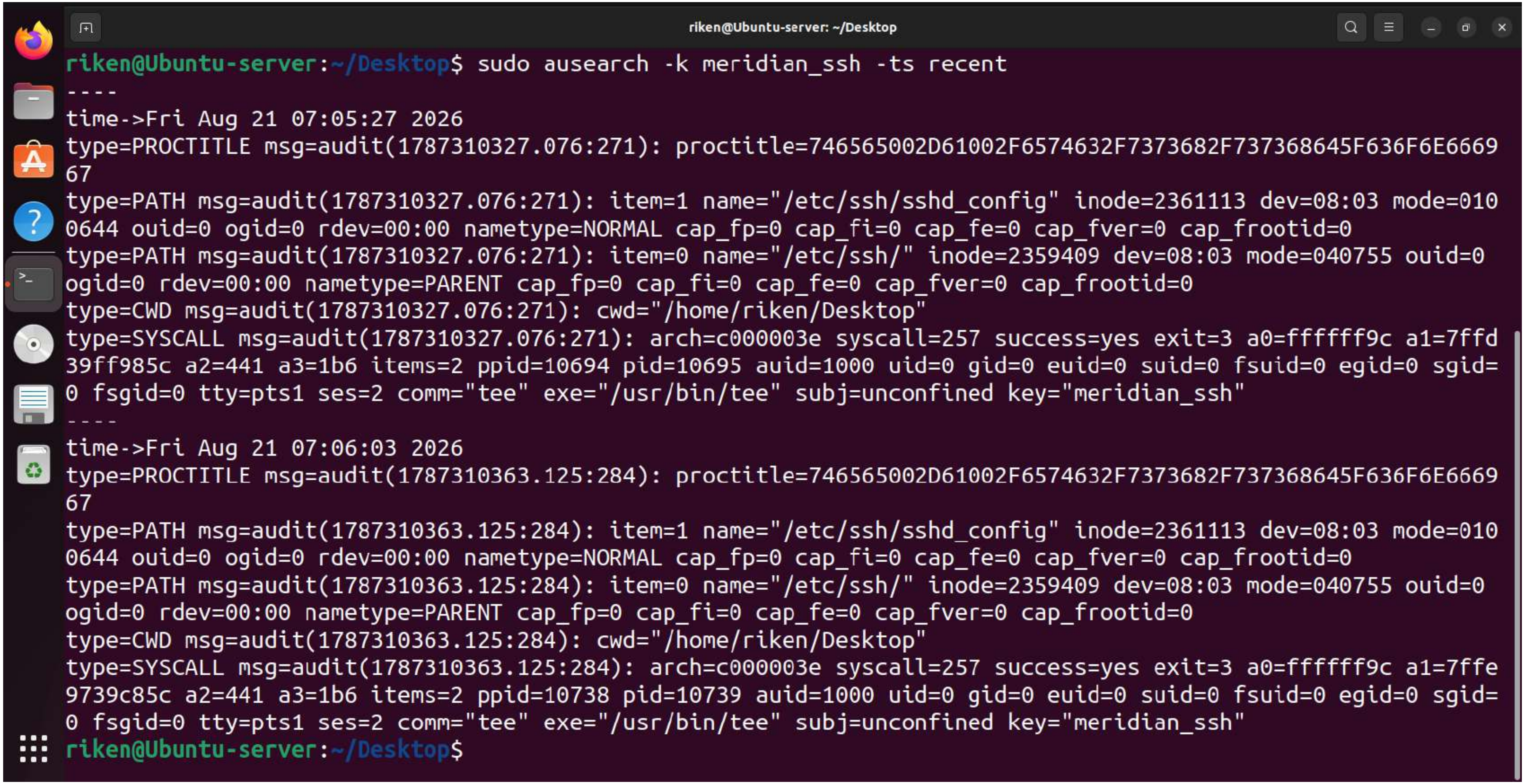

riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ sudo tail -f /var/log/suricata/eve.json | grep --line-buffered '"event_type":"alert"'
{"timestamp":"2026-08-21T06:35:05.901823-0400","flow_id":1559967183782591,"in_iface":"ens33","event_type":"alert","src_ip":"192.168.192.30","src_port":0,"dest_ip":"192.168.192.20","dest_port":0,"proto":"ICMP","icmp_type":8,"icmp_code":9,"alert":{"action":"allowed","gid":1,"signature_id":2200025,"rev":2,"signature":"SURICATA ICMPv4 unknown code","category":"Generic Protocol Command Decode","severity":3},"flow":{"pkts_toserver":1,"pkts_toclient":0,"bytes_toserver":162,"bytes_toclient":0,"start":"2026-08-21T06:35:05.901823-0400"}}
{"timestamp":"2026-08-21T06:35:05.901855-0400","flow_id":1559967183782591,"in_iface":"ens33","event_type":"alert","src_ip":"192.168.192.20","src_port":0,"dest_ip":"192.168.192.30","dest_port":0,"proto":"ICMP","icmp_type":0,"icmp_code":9,"alert":{"action":"allowed","gid":1,"signature_id":2200025,"rev":2,"signature":"SURICATA ICMPv4 unknown code","category":"Generic Protocol Command Decode","severity":3},"flow":{"pkts_toserver":1,"pkts_toclient":1,"bytes_toserver":162,"bytes_toclient":162,"start":"2026-08-21T06:35:05.901823-0400"}}
{"timestamp":"2026-08-21T06:35:05.901817-0400","flow_id":600005485904569,"in_iface":"ens33","event_type":"alert","src_ip":"192.168.192.30","src_port":0,"dest_ip":"192.168.192.20","dest_port":0,"proto":"ICMP","icmp_type":8,"icmp_code":9,"alert":{"action":"allowed","gid":1,"signature_id":2200025,"rev":2,"signature":"SURICATA ICMPv4 unknown code","category":"Generic Protocol Command Decode","severity":3},"flow":{"pkts_toserver":1,"pkts_toclient":0,"bytes_toserver":162,"bytes_toclient":0,"start":"2026-08-21T06:35:05.901817-0400"}}
{"timestamp":"2026-08-21T06:35:05.901854-0400","flow_id":600005485904569,"in_iface":"ens33","event_type":"alert","src_ip":"192.168.192.20","src_port":0,"dest_ip":"192.168.192.30","dest_port":0,"proto":"ICMP","icmp_type":0,"icmp_code":9,"alert":{"action":"allowed","gid":1,"signature_id":2200025,"rev":2,"signature":"SURICATA ICMPv4 unknown code","category":"Generic Protocol Command Decode","severity":3},"flow":{"pkts_toserver":1,"pkts_toclient":1,"bytes_toserver":162,"bytes_toclient":162,"start":"2026-08-21T06:35:05.901817-0400"}}
{"timestamp":"2026-08-21T06:35:07.438512-0400","flow_id":1559967183782591,"in_iface":"ens33","event_type":"alert","src_ip":"192.168.192.30","src_port":0,"dest_ip":"192.168.192.20","dest_port":0,"proto":"ICMP","icmp_type":8,"icmp_code":9,"alert":{"action":"allowed","gid":1,"signature_id":2200025,"rev":2,"signature":"SURICATA ICMPv4 unknown code","category":"Generic Protocol Command Decode","severity":3},"flow":{"pkts_toserver":3,

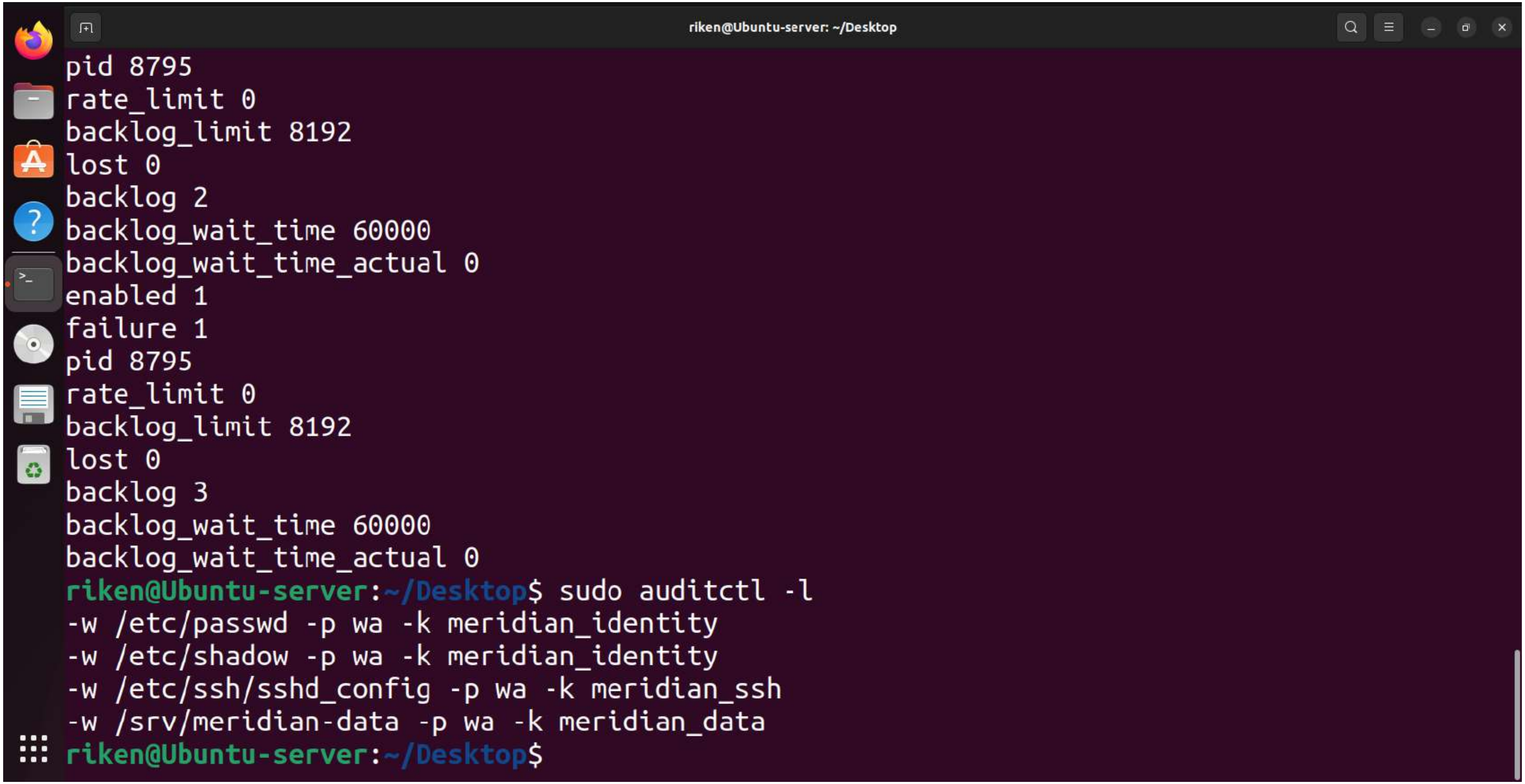

riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ suricata --build-info
This is Suricata version 6.0.4 RELEASE
Features: NFQ PCAP_SET_BUFF AF_PACKET HAVE_PACKET_FANOUT LIBCAP_NG LIBNET1.1 HAVE_HTP_URI_NORMALIZE_HOOK PCRE_JIT HAVE_NSS HAVE_LUA HAVE_LUAJIT HAVE_LIBJANSSON TLS TLS_C11 MAGIC RUST
SIMD support: none
Atomic intrinsics: 1 2 4 8 byte(s)
64-bits, Little-endian architecture
GCC version 11.2.0, C version 201112
compiled with _FORTIFY_SOURCE=2
L1 cache line size (CLS)=64
thread local storage method: _Thread_local
compiled with LibHTP v0.5.39, linked against LibHTP v0.5.39
Suricata Configuration:
AF_PACKET support: yes
eBPF support: yes
XDP support: yes
PF_RING support: no
NFQueue support: yes
NFLOG support: yes
IPFW support: no
Netmap support: no
DAG enabled: no
Napatech enabled: no
WinDivert enabled: no
Unix socket enabled: yes
Detection enabled: yes
Libmagic support: yes
libnss support: yes
libnspr support: yes
libjansson support: yes
hiredis support: yes
hiredis async with libevent: yes
Prelude support: no
PCRE jit: yes
LUA support: yes, through luajit
libluajit: yes
GeoIP2 support: yes
Non-bundled http: yes
Hyperscan support: yes
Libnet support: yes
liblz4 support: yes
HTTP2 decompression: no

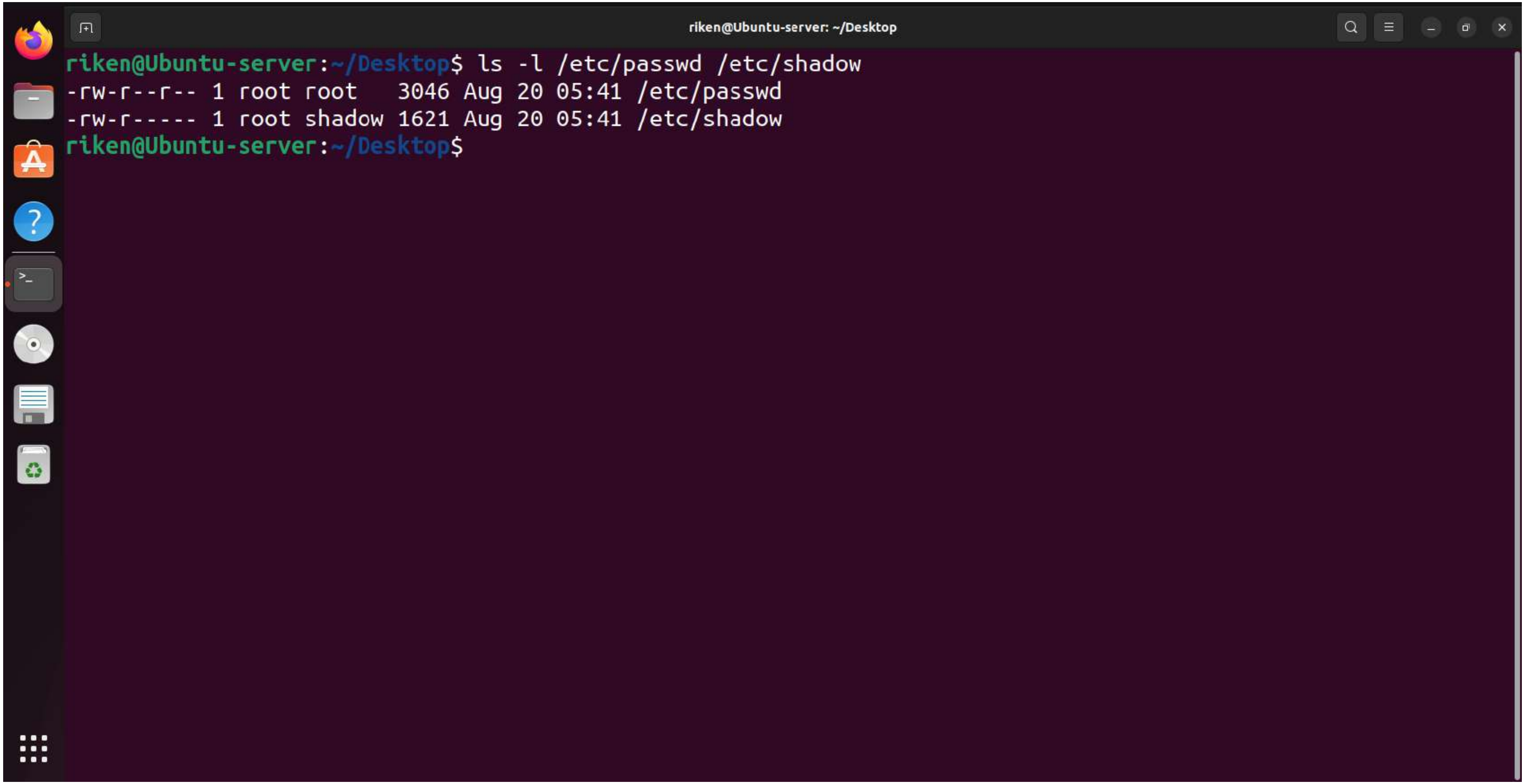

riken@Ubuntu-server: ~/Desktop
Synchronizing state of auditd.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable auditd
riken@Ubuntu-server:~/Desktop$ sudo systemctl status auditd --no-pager
● auditd.service - Security Auditing Service
Loaded: loaded (/lib/systemd/system/auditd.service; enabled; vendor preset: enabled)
Active: active (running) since Fri 2026-08-21 06:43:06 EDT; 42s ago
Docs: man:auditd(8)
https://github.com/linux-audit/audit-documentation
Main PID: 8795 (auditd)
Tasks: 2 (limit: 4540)
Memory: 528.0K
CPU: 35ms
CGroup: /system.slice/auditd.service
└─8795 /sbin/auditd
Aug 21 06:43:06 Ubuntu-server augenrules[8808]: enabled 1
Aug 21 06:43:06 Ubuntu-server augenrules[8808]: failure 1
Aug 21 06:43:06 Ubuntu-server augenrules[8808]: pid 8795
Aug 21 06:43:06 Ubuntu-server augenrules[8808]: rate_limit 0
Aug 21 06:43:06 Ubuntu-server augenrules[8808]: backlog_limit 8192
Aug 21 06:43:06 Ubuntu-server augenrules[8808]: lost 0
Aug 21 06:43:06 Ubuntu-server augenrules[8808]: backlog 4
Aug 21 06:43:06 Ubuntu-server augenrules[8808]: backlog_wait_time 60000
Aug 21 06:43:06 Ubuntu-server augenrules[8808]: backlog_wait_time_actual 0
Aug 21 06:43:06 Ubuntu-server systemd[1]: Started Security Auditing Service.
riken@Ubuntu-server:~/Desktop$

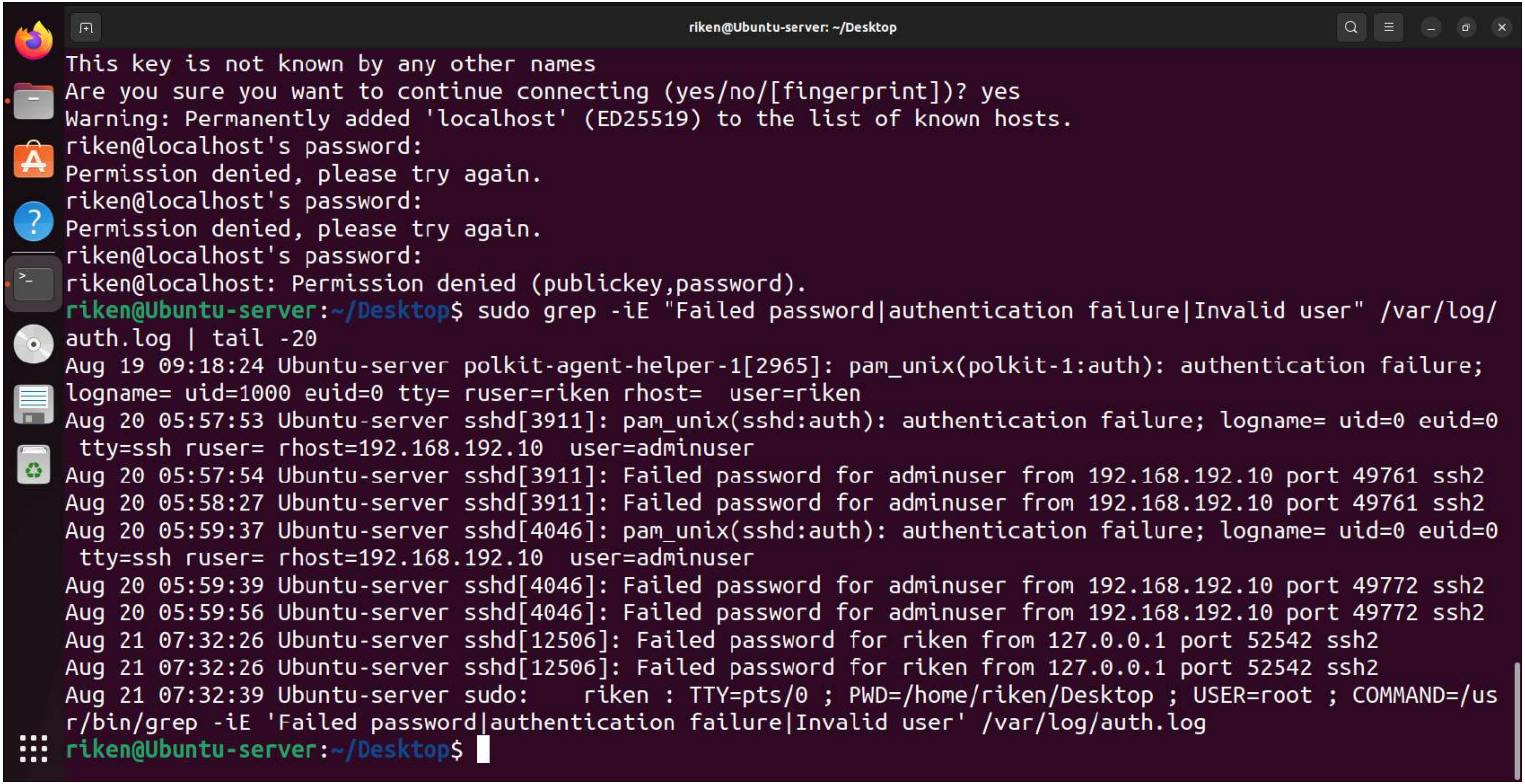
riken@Ubuntu-server: ~/Desktop

```
-w /etc/passwd -p wa -k meridian_identity
-w /etc/shadow -p wa -k meridian_identity
-w /etc/ssh/sshd_config -p wa -k meridian_ssh
-w /srv/meridian-data -p wa -k meridian_data
riken@Ubuntu-server:~/Desktop$ sudo touch /srv/meridian-data/audit-test.txt
riken@Ubuntu-server:~/Desktop$ sudo ausearch -k meridian_data -ts recent
----
time->Fri Aug 21 07:01:48 2026
type=PROCTITLE msg=audit(1787310108.368:248): proctitle=746F756368002F7372762F6D6572696469
616E2D646174612F61756469742D746573742E747874
type=PATH msg=audit(1787310108.368:248): item=1 name="/srv/meridian-data/audit-test.txt" i
node=1179652 dev=08:03 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=CREATE cap_fp=0 cap_
fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1787310108.368:248): item=0 name="/srv/meridian-data/" inode=1179650 d
ev=08:03 mode=040750 ouid=1002 ogid=1002 rdev=00:00 nametype=PARENT cap_fp=0 cap_fi=0 cap_
fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1787310108.368:248): cwd="/home/riken/Desktop"
type=SYSCALL msg=audit(1787310108.368:248): arch=c000003e syscall=257 success=yes exit=3 a
0=ffffff9c a1=7fff64514841 a2=941 a3=1b6 items=2 ppid=10403 pid=10404 auid=1000 uid=0 gid=
0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts1 ses=2 comm="touch" exe="/usr/bin/to
uch" subj=unconfined key="meridian_data"
riken@Ubuntu-server:~/Desktop$
```


riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ sudo ausearch -k meridian_ssh -ts recent
----
time->Fri Aug 21 07:05:27 2026
type=PROCTITLE msg=audit(1787310327.076:271): proctitle=746565002D61002F6574632F7373682F737368645F636F6E666967
type=PATH msg=audit(1787310327.076:271): item=1 name="/etc/ssh/sshd_config" inode=2361113 dev=08:03 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1787310327.076:271): item=0 name="/etc/ssh/" inode=2359409 dev=08:03 mode=040755 ouid=0 ogid=0 rdev=00:00 nametype=PARENT cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1787310327.076:271): cwd="/home/riken/Desktop"
type=SYSCALL msg=audit(1787310327.076:271): arch=c000003e syscall=257 success=yes exit=3 a0=ffffff9c a1=7ffd39ff985c a2=441 a3=1b6 items=2 ppid=10694 pid=10695 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts1 ses=2 comm="tee" exe="/usr/bin/tee" subj=unconfined key="meridian_ssh"
----
time->Fri Aug 21 07:06:03 2026
type=PROCTITLE msg=audit(1787310363.125:284): proctitle=746565002D61002F6574632F7373682F737368645F636F6E666967
type=PATH msg=audit(1787310363.125:284): item=1 name="/etc/ssh/sshd_config" inode=2361113 dev=08:03 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(1787310363.125:284): item=0 name="/etc/ssh/" inode=2359409 dev=08:03 mode=040755 ouid=0 ogid=0 rdev=00:00 nametype=PARENT cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1787310363.125:284): cwd="/home/riken/Desktop"
type=SYSCALL msg=audit(1787310363.125:284): arch=c000003e syscall=257 success=yes exit=3 a0=ffffff9c a1=7ffe9739c85c a2=441 a3=1b6 items=2 ppid=10738 pid=10739 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts1 ses=2 comm="tee" exe="/usr/bin/tee" subj=unconfined key="meridian_ssh"
riken@Ubuntu-server:~/Desktop$

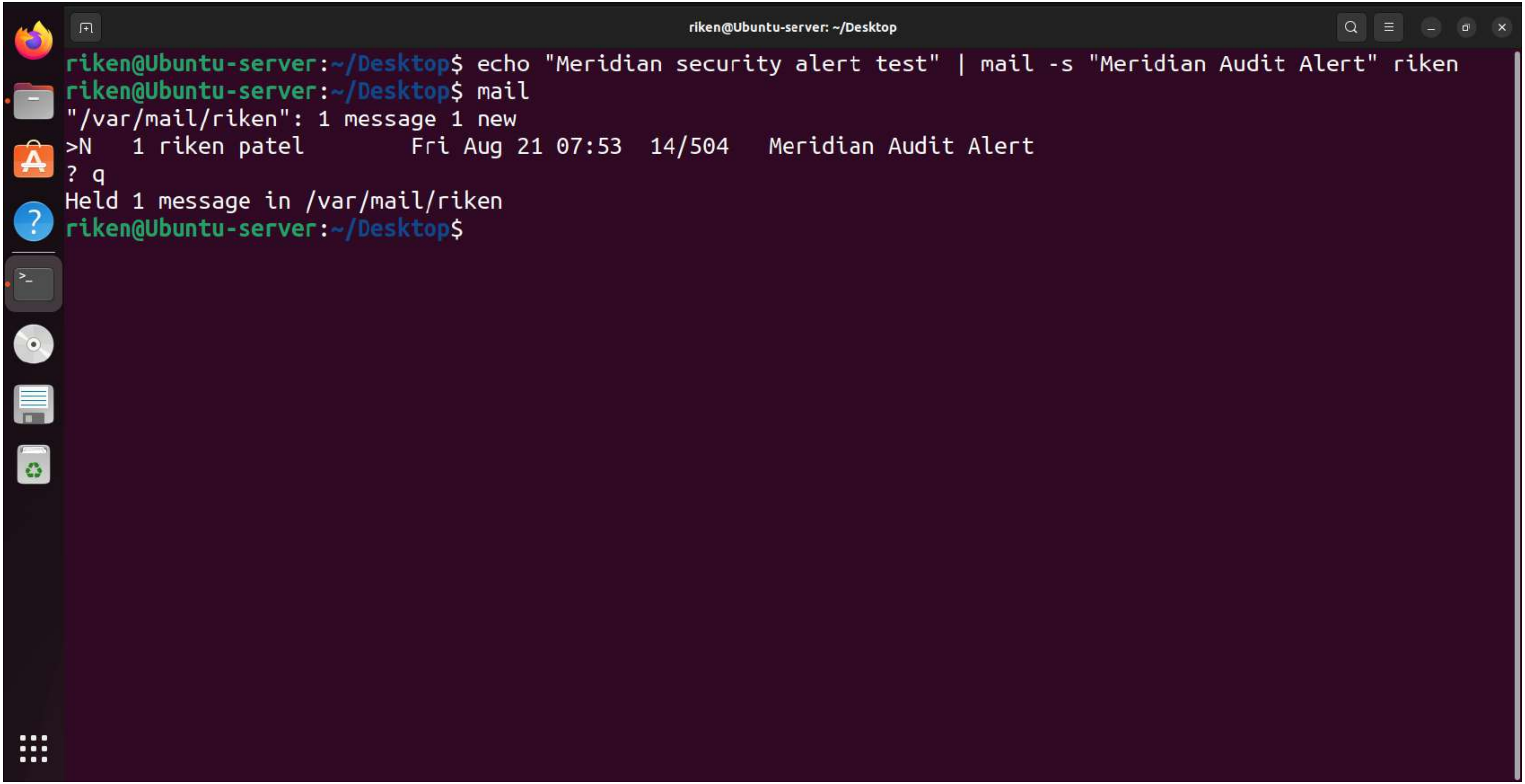

riken@Ubuntu-server: ~/Desktop
pid 8795
rate_limit 0
backlog_limit 8192
lost 0
backlog 2
backlog_wait_time 60000
backlog_wait_time_actual 0
enabled 1
failure 1
pid 8795
rate_limit 0
backlog_limit 8192
lost 0
backlog 3
backlog_wait_time 60000
backlog_wait_time_actual 0
riken@Ubuntu-server:~/Desktop$ sudo auditctl -l
-w /etc/passwd -p wa -k meridian_identity
-w /etc/shadow -p wa -k meridian_identity
-w /etc/ssh/sshd_config -p wa -k meridian_ssh
-w /srv/meridian-data -p wa -k meridian_data
riken@Ubuntu-server:~/Desktop$

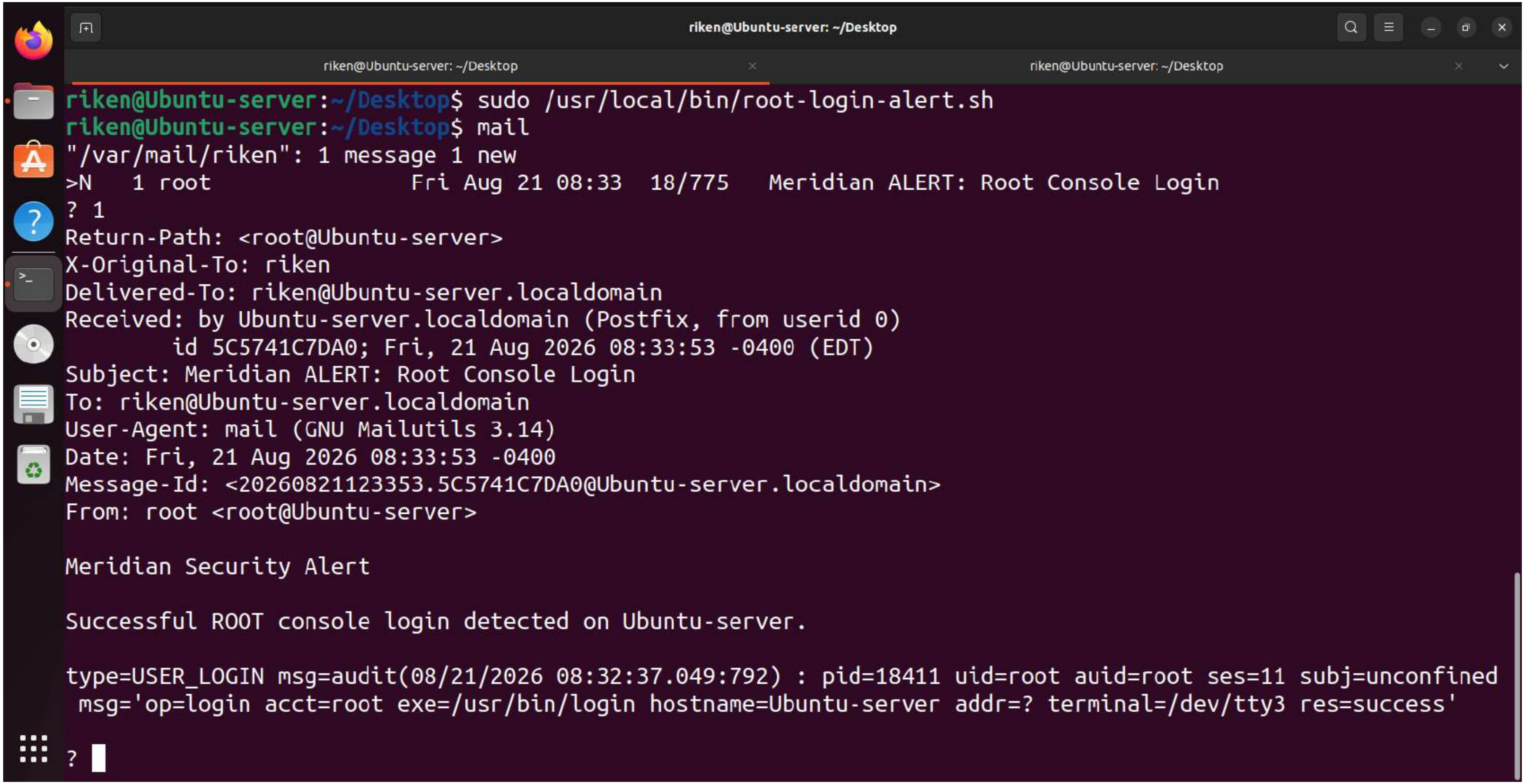

riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ ls -l /etc/passwd /etc/shadow
-rw-r--r-- 1 root root   3046 Aug 20 05:41 /etc/passwd
-rw-r----- 1 root shadow 1621 Aug 20 05:41 /etc/shadow
riken@Ubuntu-server:~/Desktop$

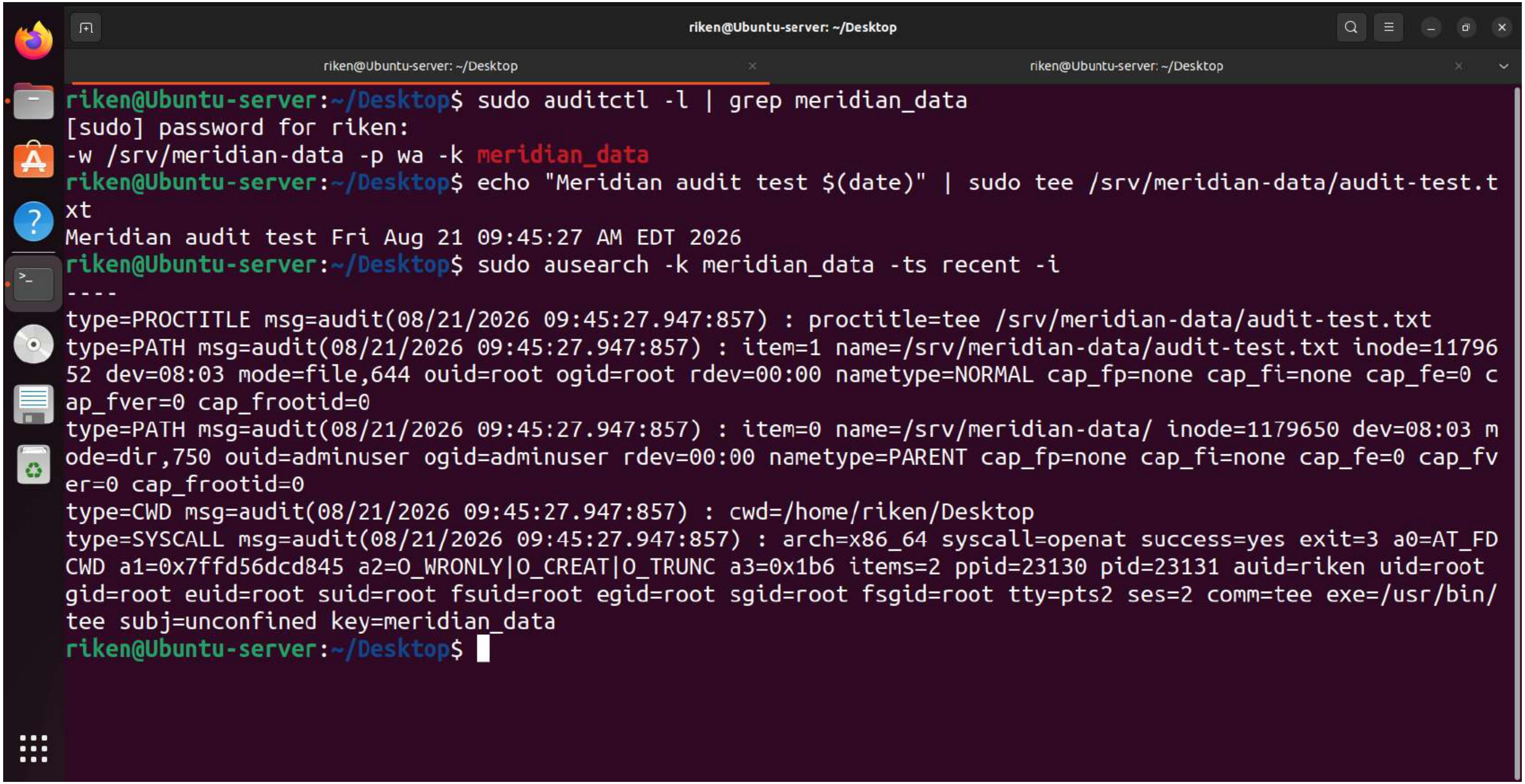

riken@Ubuntu-server: ~/Desktop
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.
riken@localhost's password:
Permission denied, please try again.
riken@localhost's password:
Permission denied, please try again.
riken@localhost's password:
riken@localhost: Permission denied (publickey,password).
riken@Ubuntu-server:~/Desktop$ sudo grep -iE "Failed password|authentication failure|Invalid user" /var/log/auth.log | tail -20
Aug 19 09:18:24 Ubuntu-server polkit-agent-helper-1[2965]: pam_unix(polkit-1:auth): authentication failure; logname= uid=1000 euid=0 tty= ruser=riken rhost= user=riken
Aug 20 05:57:53 Ubuntu-server sshd[3911]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.192.10 user=adminuser
Aug 20 05:57:54 Ubuntu-server sshd[3911]: Failed password for adminuser from 192.168.192.10 port 49761 ssh2
Aug 20 05:58:27 Ubuntu-server sshd[3911]: Failed password for adminuser from 192.168.192.10 port 49761 ssh2
Aug 20 05:59:37 Ubuntu-server sshd[4046]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.192.10 user=adminuser
Aug 20 05:59:39 Ubuntu-server sshd[4046]: Failed password for adminuser from 192.168.192.10 port 49772 ssh2
Aug 20 05:59:56 Ubuntu-server sshd[4046]: Failed password for adminuser from 192.168.192.10 port 49772 ssh2
Aug 21 07:32:26 Ubuntu-server sshd[12506]: Failed password for riken from 127.0.0.1 port 52542 ssh2
Aug 21 07:32:26 Ubuntu-server sshd[12506]: Failed password for riken from 127.0.0.1 port 52542 ssh2
Aug 21 07:32:39 Ubuntu-server sudo: riken : TTY=pts/0 ; PWD=/home/riken/Desktop ; USER=root ; COMMAND=/usr/bin/grep -iE 'Failed password|authentication failure|Invalid user' /var/log/auth.log
riken@Ubuntu-server:~/Desktop$

riken@Ubuntu-server: ~/Desktop

```
Setting up libgsasl7:amd64 (1.10.0-5) ...
Setting up libmailutils8:amd64 (1:3.14-1) ...
Setting up mailutils (1:3.14-1) ...
update-alternatives: using /usr/bin/frm.mailutils to provide /usr/bin/frm (frm) in auto mode
update-alternatives: using /usr/bin/from.mailutils to provide /usr/bin/from (from) in auto mode
update-alternatives: using /usr/bin/messages.mailutils to provide /usr/bin/messages (messages) in auto mode
update-alternatives: using /usr/bin/movemail.mailutils to provide /usr/bin/movemail (movemail) in auto mode
update-alternatives: using /usr/bin/readmsg.mailutils to provide /usr/bin/readmsg (readmsg) in auto mode
update-alternatives: using /usr/bin/dotlock.mailutils to provide /usr/bin/dotlock (dotlock) in auto mode
update-alternatives: using /usr/bin/mail.mailutils to provide /usr/bin/mailx (mailx) in auto mode
Processing triggers for rsyslog (8.2112.0-2ubuntu2.2) ...
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
riken@Ubuntu-server:~/Desktop$ sudo systemctl status postfix --no-pager
● postfix.service - Postfix Mail Transport Agent
     Loaded: loaded (/lib/systemd/system/postfix.service; enabled; vendor preset: enabled)
     Active: active (exited) since Fri 2026-08-21 07:51:17 EDT; 9s ago
       Docs: man:postfix(1)
    Process: 15425 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 15425 (code=exited, status=0/SUCCESS)
        CPU: 1ms

Aug 21 07:51:17 Ubuntu-server systemd[1]: Starting Postfix Mail Transport Agent...
Aug 21 07:51:17 Ubuntu-server systemd[1]: Finished Postfix Mail Transport Agent.
riken@Ubuntu-server:~/Desktop$
```


riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ echo "Meridian security alert test" | mail -s "Meridian Audit Alert" riken
riken@Ubuntu-server:~/Desktop$ mail
"/var/mail/riken": 1 message 1 new
>N 1 riken patel Fri Aug 21 07:53 14/504 Meridian Audit Alert
? q
Held 1 message in /var/mail/riken
riken@Ubuntu-server:~/Desktop$

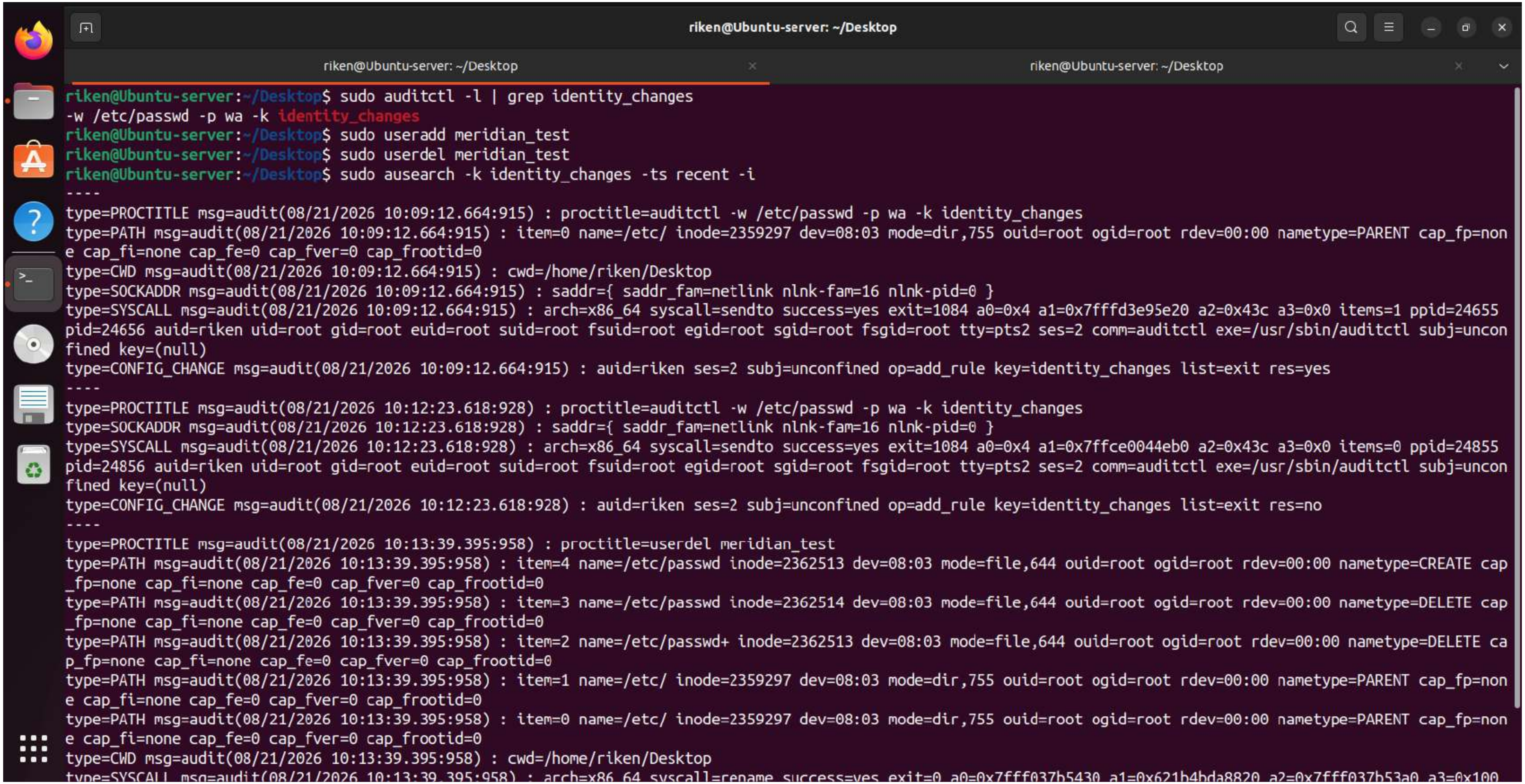

riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ sudo /usr/local/bin/root-login-alert.sh
riken@Ubuntu-server:~/Desktop$ mail
"/var/mail/riken": 1 message 1 new
>N 1 root Fri Aug 21 08:33 18/775 Meridian ALERT: Root Console Login
? 1
Return-Path: <root@Ubuntu-server>
X-Original-To: riken
Delivered-To: riken@Ubuntu-server.localdomain
Received: by Ubuntu-server.localdomain (Postfix, from userid 0)
id 5C5741C7DA0; Fri, 21 Aug 2026 08:33:53 -0400 (EDT)
Subject: Meridian ALERT: Root Console Login
To: riken@Ubuntu-server.localdomain
User-Agent: mail (GNU Mailutils 3.14)
Date: Fri, 21 Aug 2026 08:33:53 -0400
Message-Id: <20260821123353.5C5741C7DA0@Ubuntu-server.localdomain>
From: root <root@Ubuntu-server>
Meridian Security Alert
Successful ROOT console login detected on Ubuntu-server.
type=USER_LOGIN msg=audit(08/21/2026 08:32:37.049:792) : pid=18411 uid=root auid=root ses=11 subj=unconfined
msg='op=login acct=root exe=/usr/bin/login hostname=Ubuntu-server addr=? terminal=/dev/tty3 res=success'
?

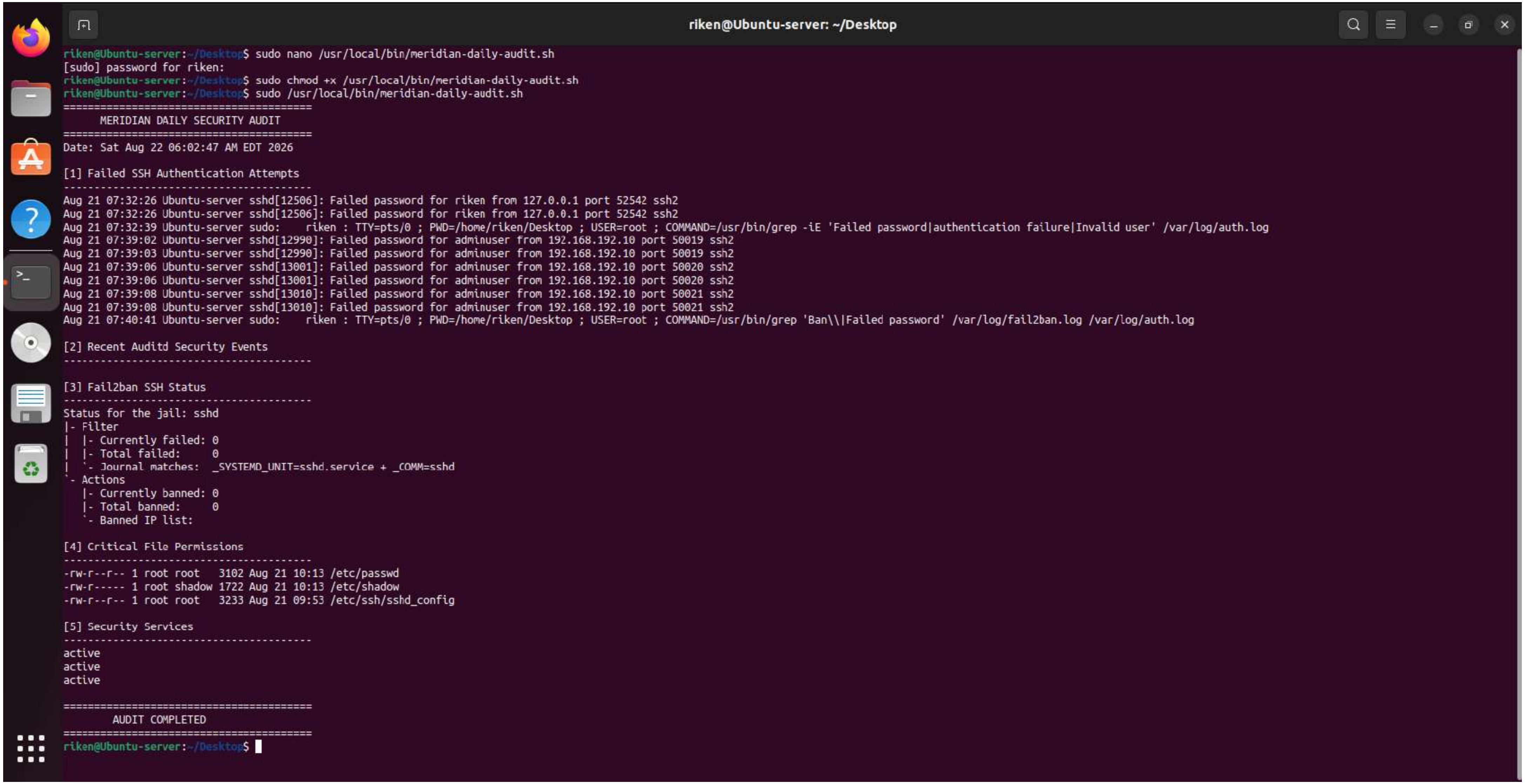

riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ sudo auditctl -l | grep meridian_data
[sudo] password for riken:
-w /srv/meridian-data -p wa -k meridian_data
riken@Ubuntu-server:~/Desktop$ echo "Meridian audit test $(date)" | sudo tee /srv/meridian-data/audit-test.txt
Meridian audit test Fri Aug 21 09:45:27 AM EDT 2026
riken@Ubuntu-server:~/Desktop$ sudo ausearch -k meridian_data -ts recent -i
----
type=PROCTITLE msg=audit(08/21/2026 09:45:27.947:857) : proctitle=tee /srv/meridian-data/audit-test.txt
type=PATH msg=audit(08/21/2026 09:45:27.947:857) : item=1 name=/srv/meridian-data/audit-test.txt inode=1179652 dev=08:03 mode=file,644 ouid=root ogid=root rdev=00:00 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(08/21/2026 09:45:27.947:857) : item=0 name=/srv/meridian-data/ inode=1179650 dev=08:03 mode=dir,750 ouid=adminuser ogid=adminuser rdev=00:00 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(08/21/2026 09:45:27.947:857) : cwd=/home/riken/Desktop
type=SYSCALL msg=audit(08/21/2026 09:45:27.947:857) : arch=x86_64 syscall=openat success=yes exit=3 a0=AT_FDCWD a1=0x7ffd56dcd845 a2=O_WRONLY|O_CREAT|O_TRUNC a3=0x1b6 items=2 ppid=23130 pid=23131 auid=riken uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=pts2 ses=2 comm=tee exe=/usr/bin/tee subj=unconfined key=meridian_data
riken@Ubuntu-server:~/Desktop$

riken@Ubuntu-server: ~/Desktop

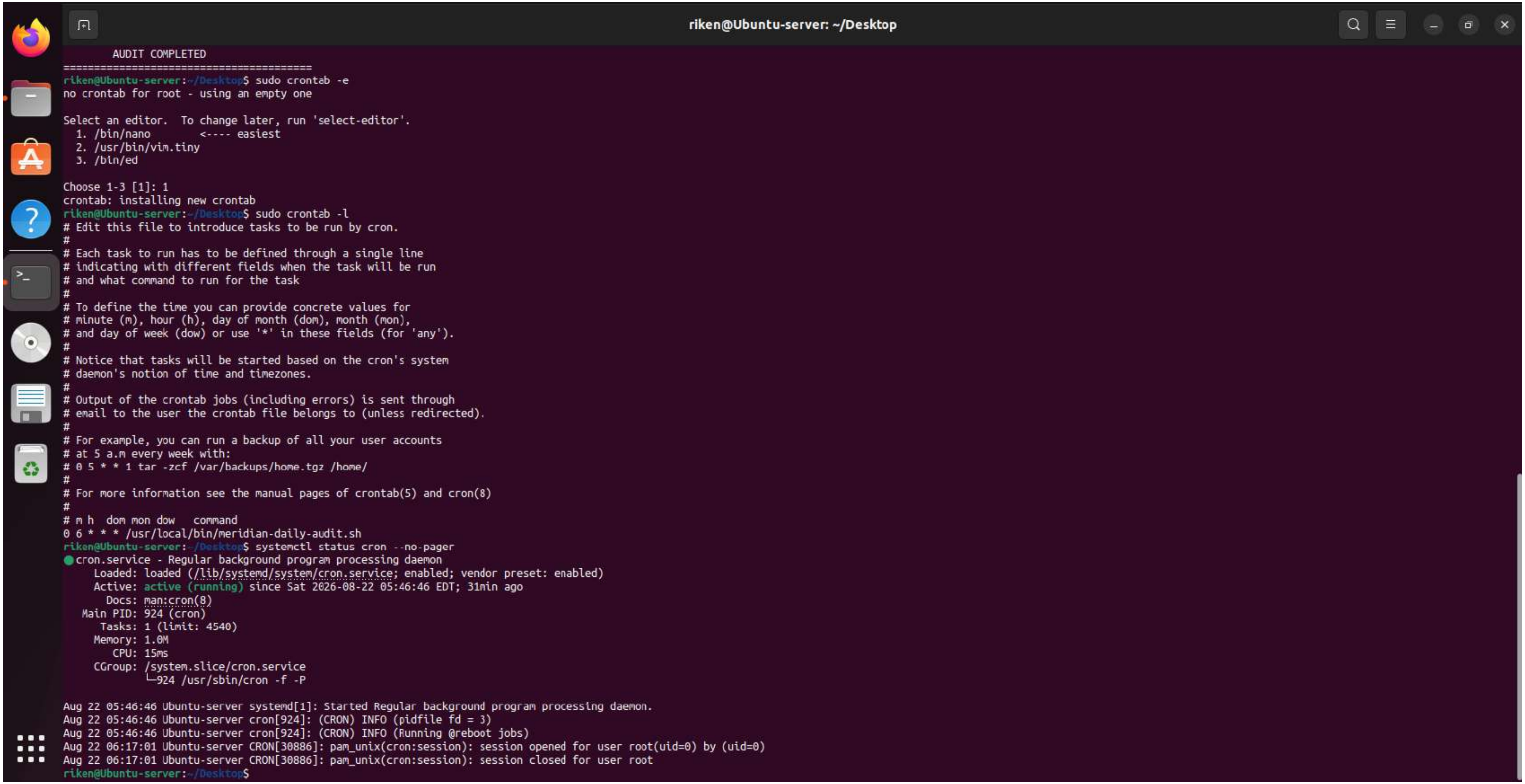
riken@Ubuntu-server: ~/Desktop | riken@Ubuntu-server: ~/Desktop

```
riken@Ubuntu-server:~/Desktop$ sudo ausearch -f /etc/ssh/sshd_config -ts recent -i
----
type=PROCTITLE msg=audit(08/21/2026 09:52:57.971:883) : proctitle=tee -a /etc/ssh/sshd_config
type=PATH msg=audit(08/21/2026 09:52:57.971:883) : item=1 name=/etc/ssh/sshd_config inode=2361113 dev=08:03
mode=file,644 ouid=root ogid=root rdev=00:00 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap
_frootid=0
type=PATH msg=audit(08/21/2026 09:52:57.971:883) : item=0 name=/etc/ssh/ inode=2359409 dev=08:03 mode=dir,75
5 ouid=root ogid=root rdev=00:00 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(08/21/2026 09:52:57.971:883) : cwd=/home/riken/Desktop
type=SYSCALL msg=audit(08/21/2026 09:52:57.971:883) : arch=x86_64 syscall=openat success=yes exit=3 a0=AT_FD
CWD a1=0x7ffc222fd85c a2=O_WRONLY|O_CREAT|O_APPEND a3=0x1b6 items=2 ppid=23607 pid=23608 auid=riken uid=root
 gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=pts2 ses=2 comm=tee exe=/usr/bin
/tee subj=unconfined key=meridian_ssh
----
type=PROCTITLE msg=audit(08/21/2026 09:53:38.903:890) : proctitle=sed -i /# Meridian audit test/d /etc/ssh/s
shd_config
type=PATH msg=audit(08/21/2026 09:53:38.903:890) : item=4 name=/etc/ssh/sshd_config inode=2362451 dev=08:03
mode=file,644 ouid=root ogid=root rdev=00:00 nametype=CREATE cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap
_frootid=0
type=PATH msg=audit(08/21/2026 09:53:38.903:890) : item=3 name=/etc/ssh/sshd_config inode=2361113 dev=08:03
mode=file,644 ouid=root ogid=root rdev=00:00 nametype=DELETE cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap
_frootid=0
type=PATH msg=audit(08/21/2026 09:53:38.903:890) : item=2 name=/etc/ssh/sedqw2Bt7 inode=2362451 dev=08:03 mo
de=file,644 ouid=root ogid=root rdev=00:00 nametype=DELETE cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_f
rootid=0
```

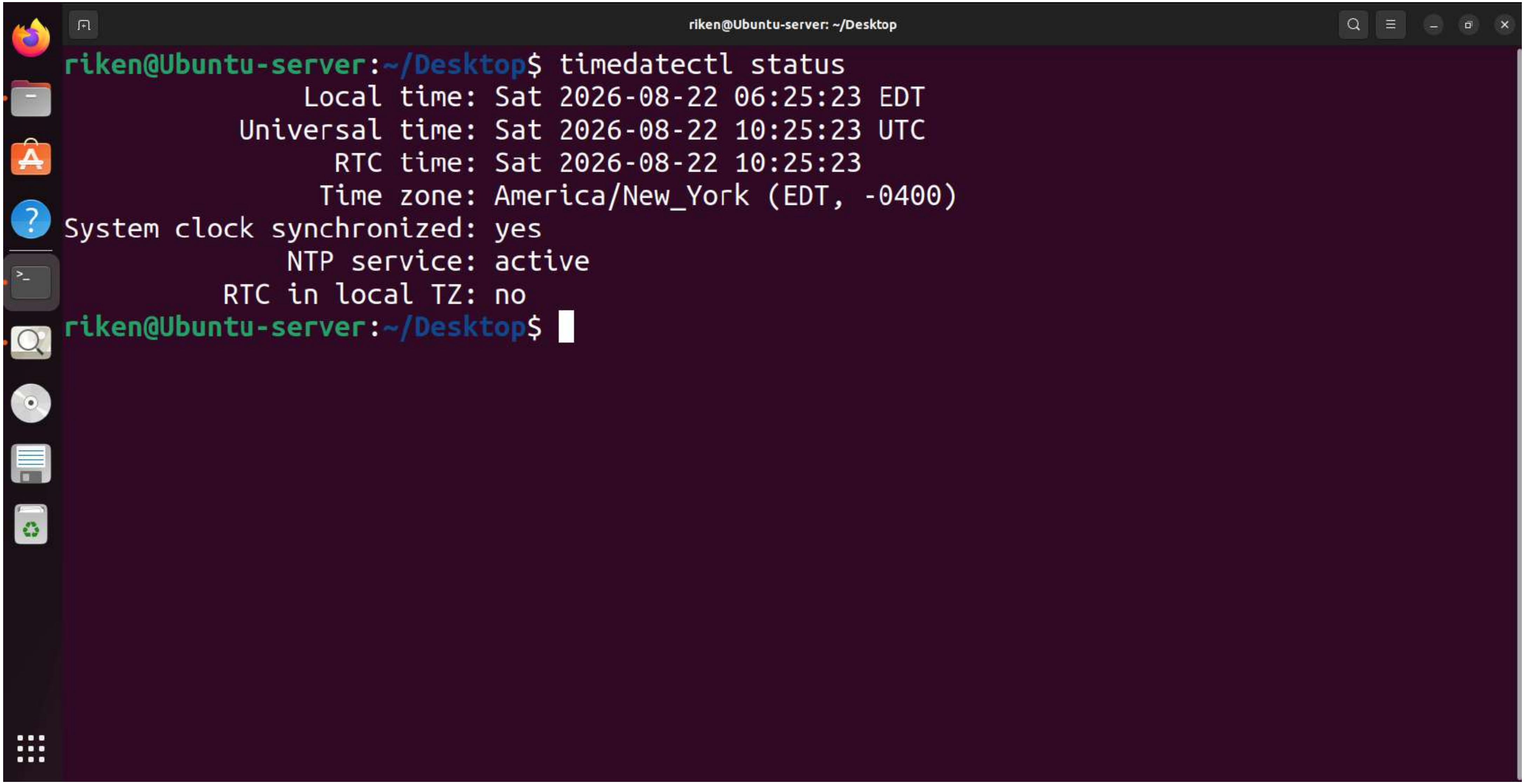

riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ sudo auditctl -l | grep identity_changes
-w /etc/passwd -p wa -k identity_changes
riken@Ubuntu-server:~/Desktop$ sudo useradd meridian_test
riken@Ubuntu-server:~/Desktop$ sudo userdel meridian_test
riken@Ubuntu-server:~/Desktop$ sudo ausearch -k identity_changes -ts recent -i
----
type=PROCTITLE msg=audit(08/21/2026 10:09:12.664:915) : proctitle=auditctl -w /etc/passwd -p wa -k identity_changes
type=PATH msg=audit(08/21/2026 10:09:12.664:915) : item=0 name=/etc/ inode=2359297 dev=08:03 mode=dir,755 ouid=root ogid=root rdev=00:00 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(08/21/2026 10:09:12.664:915) : cwd=/home/riken/Desktop
type=SOCKADDR msg=audit(08/21/2026 10:09:12.664:915) : saddr={ saddr_fam=netlink nlnk-fam=16 nlnk-pid=0 }
type=SYSCALL msg=audit(08/21/2026 10:09:12.664:915) : arch=x86_64 syscall=sendto success=yes exit=1084 a0=0x4 a1=0x7fffd3e95e20 a2=0x43c a3=0x0 items=1 ppid=24655 pid=24656 auid=riken uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=pts2 ses=2 comm=auditctl exe=/usr/sbin/auditctl subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(08/21/2026 10:09:12.664:915) : auid=riken ses=2 subj=unconfined op=add_rule key=identity_changes list=exit res=yes
----
type=PROCTITLE msg=audit(08/21/2026 10:12:23.618:928) : proctitle=auditctl -w /etc/passwd -p wa -k identity_changes
type=SOCKADDR msg=audit(08/21/2026 10:12:23.618:928) : saddr={ saddr_fam=netlink nlnk-fam=16 nlnk-pid=0 }
type=SYSCALL msg=audit(08/21/2026 10:12:23.618:928) : arch=x86_64 syscall=sendto success=yes exit=1084 a0=0x4 a1=0x7ffce0044eb0 a2=0x43c a3=0x0 items=0 ppid=24855 pid=24856 auid=riken uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=pts2 ses=2 comm=auditctl exe=/usr/sbin/auditctl subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(08/21/2026 10:12:23.618:928) : auid=riken ses=2 subj=unconfined op=add_rule key=identity_changes list=exit res=no
----
type=PROCTITLE msg=audit(08/21/2026 10:13:39.395:958) : proctitle=userdel meridian_test
type=PATH msg=audit(08/21/2026 10:13:39.395:958) : item=4 name=/etc/passwd inode=2362513 dev=08:03 mode=file,644 ouid=root ogid=root rdev=00:00 nametype=CREATE cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(08/21/2026 10:13:39.395:958) : item=3 name=/etc/passwd inode=2362514 dev=08:03 mode=file,644 ouid=root ogid=root rdev=00:00 nametype=DELETE cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(08/21/2026 10:13:39.395:958) : item=2 name=/etc/passwd+ inode=2362513 dev=08:03 mode=file,644 ouid=root ogid=root rdev=00:00 nametype=DELETE cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(08/21/2026 10:13:39.395:958) : item=1 name=/etc/ inode=2359297 dev=08:03 mode=dir,755 ouid=root ogid=root rdev=00:00 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(08/21/2026 10:13:39.395:958) : item=0 name=/etc/ inode=2359297 dev=08:03 mode=dir,755 ouid=root ogid=root rdev=00:00 nametype=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(08/21/2026 10:13:39.395:958) : cwd=/home/riken/Desktop



```
riken@Ubuntu-server:~/Desktop$ sudo nano /usr/local/bin/meridian-daily-audit.sh
[sudo] password for riken:
riken@Ubuntu-server:~/Desktop$ sudo chmod +x /usr/local/bin/meridian-daily-audit.sh
riken@Ubuntu-server:~/Desktop$ sudo /usr/local/bin/meridian-daily-audit.sh
========================================
      MERIDIAN DAILY SECURITY AUDIT
========================================
Date: Sat Aug 22 06:02:47 AM EDT 2026

[1] Failed SSH Authentication Attempts
----------------------------------------
Aug 21 07:32:26 Ubuntu-server sshd[12506]: Failed password for riken from 127.0.0.1 port 52542 ssh2
Aug 21 07:32:26 Ubuntu-server sshd[12506]: Failed password for riken from 127.0.0.1 port 52542 ssh2
Aug 21 07:32:39 Ubuntu-server sudo:    riken : TTY=pts/0 ; PWD=/home/riken/Desktop ; USER=root ; COMMAND=/usr/bin/grep -iE 'Failed password|authentication failure|Invalid user' /var/log/auth.log
Aug 21 07:39:02 Ubuntu-server sshd[12990]: Failed password for adminuser from 192.168.192.10 port 50019 ssh2
Aug 21 07:39:03 Ubuntu-server sshd[12990]: Failed password for adminuser from 192.168.192.10 port 50019 ssh2
Aug 21 07:39:06 Ubuntu-server sshd[13001]: Failed password for adminuser from 192.168.192.10 port 50020 ssh2
Aug 21 07:39:06 Ubuntu-server sshd[13001]: Failed password for adminuser from 192.168.192.10 port 50020 ssh2
Aug 21 07:39:08 Ubuntu-server sshd[13010]: Failed password for adminuser from 192.168.192.10 port 50021 ssh2
Aug 21 07:39:08 Ubuntu-server sshd[13010]: Failed password for adminuser from 192.168.192.10 port 50021 ssh2
Aug 21 07:40:41 Ubuntu-server sudo:    riken : TTY=pts/0 ; PWD=/home/riken/Desktop ; USER=root ; COMMAND=/usr/bin/grep 'Ban\\|Failed password' /var/log/fail2ban.log /var/log/auth.log

[2] Recent Auditd Security Events
----------------------------------------

[3] Fail2ban SSH Status
----------------------------------------
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  `- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:

[4] Critical File Permissions
----------------------------------------
-rw-r--r-- 1 root root   3102 Aug 21 10:13 /etc/passwd
-rw-r----- 1 root shadow 1722 Aug 21 10:13 /etc/shadow
-rw-r--r-- 1 root root   3233 Aug 21 09:53 /etc/ssh/sshd_config

[5] Security Services
----------------------------------------
active
active
active

========================================
        AUDIT COMPLETED
========================================
riken@Ubuntu-server:~/Desktop$
```

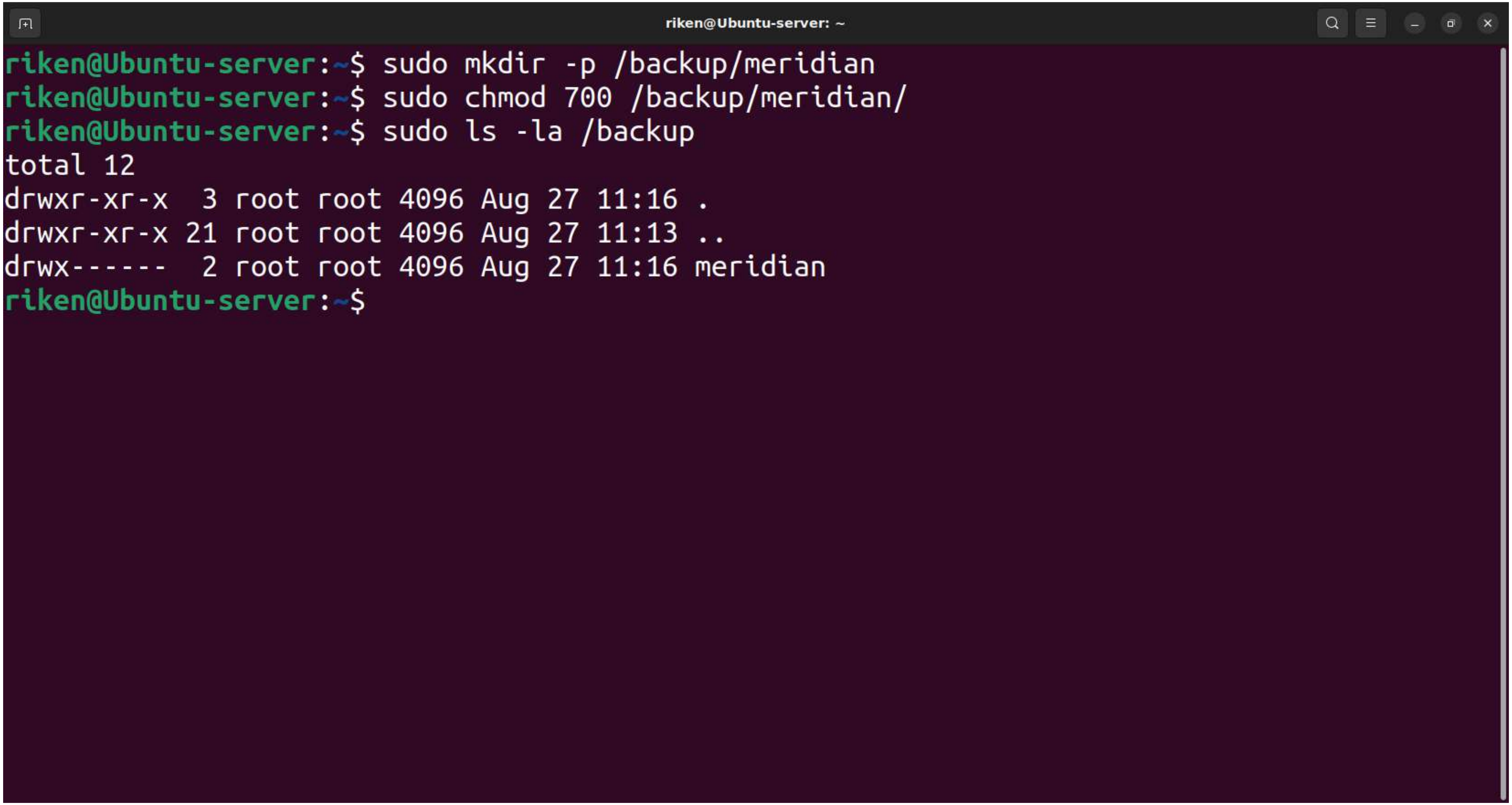

riken@Ubuntu-server: ~/Desktop

```
        AUDIT COMPLETED
========================================
riken@Ubuntu-server:~/Desktop$ sudo crontab -e
no crontab for root - using an empty one

Select an editor.  To change later, run 'select-editor'.
  1. /bin/nano        <---- easiest
  2. /usr/bin/vim.tiny
  3. /bin/ed

Choose 1-3 [1]: 1
crontab: installing new crontab
riken@Ubuntu-server:~/Desktop$ sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 6 * * * /usr/local/bin/meridian-daily-audit.sh
riken@Ubuntu-server:~/Desktop$ systemctl status cron --no-pager
● cron.service - Regular background program processing daemon
     Loaded: loaded (/lib/systemd/system/cron.service; enabled; vendor preset: enabled)
     Active: active (running) since Sat 2026-08-22 05:46:46 EDT; 31min ago
       Docs: man:cron(8)
   Main PID: 924 (cron)
      Tasks: 1 (limit: 4540)
     Memory: 1.0M
        CPU: 15ms
     CGroup: /system.slice/cron.service
             └─924 /usr/sbin/cron -f -P

Aug 22 05:46:46 Ubuntu-server systemd[1]: Started Regular background program processing daemon.
Aug 22 05:46:46 Ubuntu-server cron[924]: (CRON) INFO (pidfile fd = 3)
Aug 22 05:46:46 Ubuntu-server cron[924]: (CRON) INFO (Running @reboot jobs)
Aug 22 06:17:01 Ubuntu-server CRON[30886]: pam_unix(cron:session): session opened for user root(uid=0) by (uid=0)
Aug 22 06:17:01 Ubuntu-server CRON[30886]: pam_unix(cron:session): session closed for user root
riken@Ubuntu-server:~/Desktop$
```


riken@Ubuntu-server: ~/Desktop
riken@Ubuntu-server:~/Desktop$ timedatectl status
Local time: Sat 2026-08-22 06:25:23 EDT
Universal time: Sat 2026-08-22 10:25:23 UTC
RTC time: Sat 2026-08-22 10:25:23
Time zone: America/New_York (EDT, -0400)
System clock synchronized: yes
NTP service: active
RTC in local TZ: no
riken@Ubuntu-server:~/Desktop$

riken@Ubuntu-server: ~/Desktop

```
riken@Ubuntu-server:~/Desktop$ sudo cat /etc/logrotate.d/rsyslog
/var/log/syslog
/var/log/mail.info
/var/log/mail.warn
/var/log/mail.err
/var/log/mail.log
/var/log/daemon.log
/var/log/kern.log
/var/log/auth.log
/var/log/user.log
/var/log/lpr.log
/var/log/cron.log
/var/log/debug
/var/log/messages
{
        rotate 4
        weekly
        missingok
        notifempty
        compress
        delaycompress
        sharedscripts
        postrotate
                /usr/lib/rsyslog/rsyslog-rotate
        endscript
}
riken@Ubuntu-server:~/Desktop$ sudo systemctl status auditd --no-pager
```


riken@Ubuntu-server: ~
riken@Ubuntu-server:~$ sudo mkdir -p /backup/meridian
riken@Ubuntu-server:~$ sudo chmod 700 /backup/meridian/
riken@Ubuntu-server:~$ sudo ls -la /backup
total 12
drwxr-xr-x  3 root root 4096 Aug 27 11:16 .
drwxr-xr-x 21 root root 4096 Aug 27 11:13 ..
drwx------  2 root root 4096 Aug 27 11:16 meridian
riken@Ubuntu-server:~$

riken@Ubuntu-server: ~/Desktop

```
riken@Ubuntu-server:~/Desktop$ echo "Meridian backup test data" | sudo tee /opt/m
eridian-data/test.txt
[sudo] password for riken:
Meridian backup test data
riken@Ubuntu-server:~/Desktop$ sudo ls -l /opt/meridian-data/
total 4
-rw-r--r-- 1 root root 26 Aug 27 11:31 test.txt
riken@Ubuntu-server:~/Desktop$
```

riken@Ubuntu-server: ~

```
/meridian-data
  422  sudo ls -lh /backup/meridian/
  423  history
riken@Ubuntu-server:~$ sudo mkdir -p /tmp/meridian-restore-test
riken@Ubuntu-server:~$ sudo ls -lh /backup/meridian/
total 8.0K
-rw-r--r-- 1 root root 179 Aug 27 11:52 manual-test-backup.tar.gz
-rw-r--r-- 1 root root 179 Aug 27 11:35 meridian-backup.tar.gz
riken@Ubuntu-server:~$ sudo tar -xzf /backup/meridian/manual-test-ba
ckup.tar.gz -C /tmp/meridian-restore-test
riken@Ubuntu-server:~$ sudo ls -l /tmp/meridian-restore-test/opt/mer
idian-data/
total 4
-rw-r--r-- 1 root root 26 Aug 27 11:31 test.txt
riken@Ubuntu-server:~$ 
```

riken@Ubuntu-server: ~

```
riken@Ubuntu-server:~$ sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 6 * * * /usr/local/bin/meridian-daily-audit.sh
0 * * * * tar -czf /backup/meridian/meridian-backup-$(date +\%Y\%m\%d-\%H\%M).tar.gz /opt/meridian-data
riken@Ubuntu-server:~$
```

adminuser@Ubuntu-server: ~

```
riken@Ubuntu-server:~/Desktop$ sudo touch /tmp/meridian-priv-test
[sudo] password for riken:
riken@Ubuntu-server:~/Desktop$ sudo chown root:root /tmp/meridian-priv-test
riken@Ubuntu-server:~/Desktop$ sudo chmod 600 /tmp/meridian-priv-test
riken@Ubuntu-server:~/Desktop$ ls -l /tmp/meridian-priv-test
-rw------- 1 root root 0 Aug 28 12:01 /tmp/meridian-priv-test
riken@Ubuntu-server:~/Desktop$ su - adminuser
Password:
adminuser@Ubuntu-server:~$ whoami
adminuser
adminuser@Ubuntu-server:~$ sudo -l
[sudo] password for adminuser:
Sorry, try again.
[sudo] password for adminuser:
Sorry, user adminuser may not run sudo on Ubuntu-server.
adminuser@Ubuntu-server:~$ useradd meridian-test-user
useradd: Permission denied.
useradd: cannot lock /etc/passwd; try again later.
adminuser@Ubuntu-server:~$ chmod 777 /tmp/meridian-priv-test
chmod: changing permissions of '/tmp/meridian-priv-test': Operation not permitted
adminuser@Ubuntu-server:~$ chown adminuser:adminuser /tmp/meridian-priv-test
chown: changing ownership of '/tmp/meridian-priv-test': Operation not permitted
adminuser@Ubuntu-server:~$ touch /etc/meridian-priv-test
touch: cannot touch '/etc/meridian-priv-test': Permission denied
adminuser@Ubuntu-server:~$ crontab -l
no crontab for adminuser
adminuser@Ubuntu-server:~$ at -l
```

adminuser@Ubuntu-server: ~

```
riken@Ubuntu-server:~/Desktop$ su - adminuser
Password:
adminuser@Ubuntu-server:~$ at now
warning: commands will be executed using /bin/sh
at Sun Aug 30 21:53:00 2026
at> echo "Meridian AT job executed" > /tmp/meridian-at-test
at> <EOT>
job 4 at Sun Aug 30 21:53:00 2026
adminuser@Ubuntu-server:~$ cat /tmp/meridian-at-test
Meridian AT job executed
adminuser@Ubuntu-server:~$ atq
adminuser@Ubuntu-server:~$
```